

NORDIC TRIP/NBG-19-01; SWEB CG 19-01

**A TRANSLATIONAL RANDOMIZED PHASE III STUDY EXPLORING THE EFFECT
OF THE ADDITION OF CAPECITABINE TO CARBOPLATINUM BASED
CHEMOTHERAPY IN EARLY “TRIPLE NEGATIVE” BREAST CANCER**

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Sponsor: Department of Hematology, Oncology and Radiation Physics, Skåne University Hospital

National organisations:

Swedish Breast Cancer Group (SweBCG) and Swedish Association of Breast Oncologists (SABO)

Danish Breast cancer Group (DBCG)

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SIGNATURE PAGE

Title: **NORDIC TRIP/NBG-1; A TRANSLATIONAL RANDOMIZED PHASE III STUDY EXPLORING THE EFFECT OF THE ADDITION OF CAPECITABINE TO CARBOPLATINUM BASED CHEMOTHERAPY IN EARLY “TRIPLE NEGATIVE” BREAST CANCER**

Protocol ID no: NordicTrip / NBG-1

EudraCT no: 2018-002080-25

I hereby declare that I will conduct the study in compliance with the Protocol, ICH-GCP and all applicable regulatory requirements:

To be signed by Principal Investigator and Study Statistician.

Name	Title	Role	Signature	Date

PROTOCOL SYNOPSIS

A translational randomized phase III study exploring the effect of the addition of capecitabine to carboplatin based chemotherapy in early “triple negative” breast cancer

Sponsor	Skåne University Hospital
Phase and study type	Phase III, translational
Investigational Medical Product (IMP) (including active comparator) :	Capecitabine, carboplatin, cyclophosphamide, epirubicin, paklitaxel.
Participating countries:	Sweden, Denmark, Finland, Iceland
Participating centers	15 sites in Sweden, 8 sites in Denmark, 1 site in Finland, 1 site in Iceland
Study Period:	Estimated date of first patient enrolled: 1 December 2019 Anticipated recruitment period: 3 years Estimated date of last patient completed (treatment phase): 30 June 2023
Treatment Duration:	20-23 weeks
Follow-up:	10 years
Objectives	Primary objectives: • To compare the effect on pathologic complete response (pCR) rate of adding capecitabine to carboplatin based preoperative chemotherapy in early ER-negative and HER2-negative breast cancer. Secondary objectives: • To compare invasive disease free (IDFS), breast cancer specific survival (BCSS), distant recurrencefree survival (DRFS), and overall survival (OS). • To evaluate the tolerability defined by toxicity and dose intensity.

Translational objectives:

- To determine the pCR rates in different treatment arms stratified for Homologous Repair Deficiency (HRD) positive vs. HRD-negative/HRD-intermediate.
- To characterize different subsets of TNBC in terms of morphology, epigenetic alterations as well as somatic and inherited genetic alterations.
- To determine the pCR rate and long-term outcome in subsets of TNBC with defined molecular genetic alterations including BRCA1, BRCA2 and PALB2 germline mutations and others, BRCA1, BRCA2 and PALB2 somatic mutations and others and BRCA1 or RAD51 promoter methylation.
- To determine the pCR rate and long-term outcome in different histological subtypes of TNBC
- To determine the pCR rate and long-term outcome in subsets defined based on markers of immune response, eg tumor infiltrating lymphocytes and PDL1-expression.
- Analysis of circulating tumor DNA and circulating immune-markers as a marker of treatment response and long-term prognosis.

Endpoints:

Primary endpoint: pCR rate

Primary Translational Endpoint: pCR rate in the different study arms stratified for HRD status

Secondary endpoints: IDFS, OS, BCSS, DRFS, toxicity and outcome correlated with potential predictive biomarkers of response and toxicity.

Study Design:

Two armed, open label, multicenter.

Study treatment arms

A: **ddEC x 4 → PK x 4**, ie (epirubicin 90 mg/m² + cyclophosphamide 600 mg/m²) q2w x 4 followed by (paklitaxel 80 mg/m² day 1, 8, 15 + carboplatin AUC 5 day 1) q3w x 4.

B: **CEX x 4 → PK x 4**, ie (epirubicin 75 mg/m² + cyclophosphamide 600 mg/m² + capecitabine 1800 mg/m² day 1-14) q3w x 4 followed by (paklitaxel 80 mg/m² day 1, 8, 15 + carboplatin AUC 5 day 1) q3w x 4.

Inclusion Criteria:

Patients must meet all of the following criteria to be eligible:

1. Signed written informed consent approved by the Ethical Review Board (IRB).
2. Age 18 to 75 years.
3. Histologically confirmed unilateral adenocarcinoma of the breast where neoadjuvant chemotherapy followed by definitive surgery is planned.

4. Node positive disease (N1-3) or if clinically N0 Tumor size ≥ 20 mm.
5. ER negative tumor defined by at least one the following:
 - a. ER < 1% cells positive by IHC
 - b. ER $\leq 10\%$ cells positive by IHC and PgR $\leq 10\%$ cells positive by IHC
6. HER2-normal tumor defined by at least one of the following:
 - a. HER2 0 or 1+ by immunohistochemistry
 - b. HER2 2+ by IHC and HER2 to centromere 17 ratio < 2.0 and/or less than 6 copies by in situ hybridization (ISH).
Comment: Patients with 3+ IHC for HER2 will not be included irrespective of the outcome of ISH-analysis.
7. Consent for germline mutation screening for BRCA1, BRCA2 and other inherited breast cancer associated genes.
8. WHO performance status 0 or 1.
9. Negative pregnancy test in women of childbearing potential (premenopausal or <12 months of amenorrhea post-menopause and who have not undergone surgical sterilization).
10. Willingness of female patients of childbearing potential, male patients and their sexual partners to use an effective means of contraception during the treatment period and at least 6 months thereafter.
11. Willingness by the patient to undergo treatment and study related procedures according to the protocol.

Exclusion Criteria

Patients meeting any of the following criteria are not eligible:

1. Clinical or radiological signs of metastatic disease.
2. Clinically inflammatory breast cancer.
3. History of other malignancy within the last 5 years, except for carcinoma in situ of the cervix or non-melanoma skin cancer.
4. Previous chemotherapy for cancer or other malignant disease.
5. Charlson comorbidity index, excluding score for malignancy: (CCI) > 2,
Comment: In patients 70-75 a CCI = 3 is allowed, see appendix B.
6. Inadequate organ function, suggested by the following laboratory results:
 - a. Absolute neutrophil count < $1,5 \times 10^9/L$
 - b. Platelet count < $100 \times 10^9/L$
 - c. Haemoglobin < 90 g/L
 - d. Total bilirubin greater than the upper limit of normal (ULN) unless the patient has documented Gilbert's syndrome
 - e. ASAT (SGOT) and/or ALAT (SGPT) > $2,5 \times ULN$
 - f. ASAT (SGOT) and/or ALAT (SGPT) > $1,5 \times ULN$ with concurrent serum alkaline phosphatase (ALP) > $2,5 \times ULN$
 - g. Serum creatinine clearance < 50 ml/min
7. Concurrent peripheral neuropathy of grade 3 or greater (NCI-CTCAE, Version 5.0).
8. Patient who is actively breast feeding.

9. Assessed by the Investigator to be unable or unwilling to comply with the requirements of the protocol.

Primary aim: Pathological complete response rate after preoperative chemotherapy is the primary end-point of the study, which will be evaluated by comparing the effects of neoadjuvant administration of a carboplatin-based treatment (A) and treatment adding capecitabine (B) on pCR. By assuming a pCR rate of 52% (Poggio F., Bruzzone M. et al. 2018) in arm A, a sample size of 808 patients, 404 in each arm, is required to have 80% power to detect a difference of 10 percentage units, i.e. a pCR rate of 62% in arm B, using a two-sided Chi-squared test with the significance level set at 5%. To account for an expected low fraction of non-evaluated patients, the sample size is set at 820. Patients that are included in the study and receive study treatment, but do not undergo surgery will be evaluated as having a non-pCR.

Primary translational aim: To investigate if the effects of the treatments depend on HRD-status. More specifically, the aim is to test for differential effect of the two treatments on pCR for HRD-negative (HRD low and intermediate by oncoscan) and HRD-positive (HRD high by oncoscan) patients.

Assuming that

- a. 59% of the patients are HRD-positive (implying 41% HRD-negative)
- b. that the pCR rates in arm A and B are 52% and 62%, respectively, and
- c. that the effects of the two treatments are equal (same odds of pCR; OR=1.00) for HRD-positive patients

the OR for B vs A in the HRD-negative group can be calculated for different pCR rates for HRD-positive patients in treatment group A.

Below two examples:

- 63% pCR in HRD+ (for both arms) implies 36.2% pCR in HRD- for arm A and 60.6% pCR in HRD- for arm B corresponding to an OR of 2.71 for B vs A in HRD-. This differential effect of treatment on pCR rates in HRD+ and HRD- can be detected with 92% power with a sample size of 800 patients using a two-sided test for interaction with significance level set to 0.05.
- 67% pCR in HRD+ (for both arms) implies 30.4% pCR in HRD- for arm A and 54.8% pCR in HRD- for arm B corresponding to an OR of 2.77 for B vs A in HRD-. Also this differential effect can be detected with 92% power with a sample size of 800 patients using a two-sided test for interaction with significance level set to 0.05.

Randomization will be performed 1 : 1 between the two treatment arms A and B, with stratification for the country of inclusion, N-status and T-status.

Efficacy Assessments:

Pathological response rate in the breast and axilla, with a disappearance of all signs of invasive cancer. Cases with residual in situ deposits in the breast are included in the definition of pCR. Relapse, new cancer, survival in all patients. Quality of life will be investigated in a subset of included patients.

Safety Assessments:

Breast imaging at baseline and after 4-5 weeks of treatment, in order to exclude on-treatment disease progression.

Reporting of adverse events, biochemistry, hematology, vital signs and performance status.

An evaluation and analysis of the feasibility of the prescribed doses and pCR rate irrespective of treatment allocation will be performed after the treatment of 60 patients.

Data Management

Data will be collected in a database run by the Danish Breast Cancer Group (DBCG).

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List of Abbreviations and Definitions of Terms

Abbreviation	Explanation
AE	Adverse Event
ALAT (SGPT)	Alanine Aminotransferase (Serum Glutamic Pyruvic Transaminase)
ASAT (SGOT)	Aspartate Aminotransferase (Serum Glutamic Oxaloacetic Transaminase)
AUC	Area Under the Curve
BCSS	Breast Cancer specific Survival
BRCA	Breast Cancer Gene
CCI	Charlson Comorbidity Index
ClinR	Clinical Response
CR	Complete Remission
CRF	Case Report Form (electronic/paper)
CSA	Clinical Study Agreement
CT	Computer Tomography
CTCAE	Common Terminology Criteria for Adverse Event
DBCG	The Danish Breast Cancer Group
ddEC	Dose Dense EC (Epirubicine and Cyclophosphamide)
DRFS	Distant Recurrence Free Survival
EBCTCG	Early Breast Cancer Trialists Collaborative Group
EC	Epirubicine and Cyclophosphamide
ECG	Electrocardiogram
ER	Estrogen Receptor
GCP	Good Clinical Practice
G-CSF	Granulocyte Colony Stimulating Factor
HER2	Human Epidermal Growth Factor Receptor
HRD	Homologus repair deficiency
ICF	Informed Consent Form
ICH	International Conference on Harmonization
IDFS	Invasive Disease Free Survival
IHC	Immunohistochemistry
IMP	Investigational Medicinal Product
IND	Investigational New Drug
ISH	In Situ Hybridization
MRI	Magnetic Resonance Imaging
OS	Overall Survival
pCR	Pathological Complete Response
PD	Progressive Disease
PgR	Progesterone Receptor
PR	Partial Remission
q3w	Every 3 weeks
q2w	Every 2 weeks
RNA	Ribonucleic Acid
SAE	Serious Adverse Event
SD	Stable Disease
SOP	Standard Operating Procedure
SPC	Summary of Product Characteristics
SUSAR	Suspected Unexpected Serious Adverse Reaction
ULN	Upper Limit of Normal

1 INTRODUCTION

1.1 Background – Early breast cancer

1.1.1 Adjuvant and preoperative chemotherapy in early breast cancer

In early breast cancer disease, the administration of adjuvant chemotherapy significantly reduces the risk of recurrence and breast cancer-specific death (Peto, Davies et al. 2012). Adjuvant polychemotherapy typically leads to a one-third reduction of breast cancer-specific mortality at ten years. With the exception of HER2-positive disease, the chemotherapy regimens administered do not include targeted treatment agents, but instead depends on the general chemosensitivity of micrometastatic cancer cells. Shortly after the year 2000, when the gene-expression based “intrinsic” subtypes of breast cancer were first presented by Sorlie and colleagues, the need to sub-classify breast cancer into distinct entities in order to better understand the biological and clinical differences between breast cancers has become increasingly evident (Perou et al. 2000; Sorlie et al. 2001). The two subtypes with the least beneficial prognosis are the “HER2-enriched” and “basal-like” groups. The latter subset has a considerable overlap with so-called triple negative breast cancers (TNBC), i.e. cancers that express neither the estrogen or progesterone receptors nor overexpressing HER2. The adjuvant treatment and outcomes for patients with HER2 and ER-positive breast cancers has undergone a substantial improvement during recent decades since the advent of anti-HER2 antibody-based treatment, the more generous use of chemotherapy in ER-positive disease, and a more rational use of endocrine treatment. In TNBC a similarly positive development is still eagerly awaited. In a recent meta-analysis, the addition of carboplatin to conventional chemotherapy has been shown to lead to an increased rate of pathological complete response after preoperative treatment. However, the long-term effect regarding event-free and overall survival with the addition of carboplatin is still unclear (Poggio, Bruzzone et al. 2018).

1.1.2 Triple negative breast cancer (TNBC)

TNBC is characterized by the absence of expression of estrogen and progesterone receptors and the lack of signs of amplification of the oncogene HER2. TNBCs constitute about 8% of new breast cancers in Sweden, i.e. more than 600 cases yearly in Sweden, and an estimated incidence and mortality of more than 1500 and up to 500 cases per year, respectively, in the Nordic countries (Sweden, Denmark, Norway, Finland and Iceland). Apart from the absence of markers of endocrine sensitivity and HER2-positivity, TNBC remains a poorly defined and heterogeneous subset of breast cancers with the most adverse prognosis of all breast cancers. In the absence of treatable targets, chemotherapy and potentially bisphosphonates remain the sole medical intervention with a proven effect on long-term survival in TNBC. Although there has been some indications of benefit of the anti-VEGF antibody bevacizumab, the long-term effect of this costly drug has been a disappointment. Recent results indicate that immunotherapy may have a role in TNBC (Schmid, Adams et al. 2018) and investigations on the effect are under way.

A recent metaanalysis strongly suggests a benefit in terms of an increased pathologic complete response rate in unselected TNBC adding carboplatin to preoperative chemotherapy (Poggio, Bruzzone et al. 2018). Furthermore, subgroup analyses from four large randomized clinical trials suggest a selective benefit of the addition of capecitabine to standard adjuvant chemotherapy in this subtype.

There is strong evidence to suggest that the biology and the effect of chemotherapy differs between subsets of TNBCs both in general and with regard to response to specific agents (Rakha, Elsheikh et al. 2009). Histopathological subtypes among TNBCs include high-grade invasive carcinoma of no special type, medullary and atypical medullary cancers and metaplastic cancer; and among low-grade cancers, adenoid cystic, low-grade adenosquamous, and low-grade spindle carcinomas (Dawson, Provenzano et al. 2009). Among the intrinsic subtypes, most TNBCs fall into the basal-like cluster (Sorlie, Perou et al. 2001). Furthermore, gene expression-based subtyping within the triple negative cohort shows that it may be subdivided into four different subtypes (Brian, Lehmann, et al. 2016; Prat, Adamo 2013).

Subsets of TNBC have a defective DNA-repair. Most notably, BRCA1-associated breast cancer typically displays a triple negative histopathological phenotype and basal-like gene expression pattern, and does

indeed display defective DNA-repair as illustrated by heightened responsiveness to PARP-inhibition in these cancers, both in vitro (Farmer, McCabe et al. 2005) and clinically (Tutt, Robson et al. 2010; Robson, Im, et al. 2018; Litton, Hope et al. 2018). Also, breast cancers with a somatic mutation in BRCA1 or an epigenetic inactivation of BRCA1 typically displays a triple negative phenotype (Winter, Nilsson et al 2016; Hedenfalk, Duggan et al. 2001). Data from the population-based Swedish SCAN-B study indicate that in TNBC, 59% display a HRD-high (positive) phenotype, with 41% being either HRD-low (36%), or HRD-intermediate (5%), i.e. HRD-negative. In this study, that included 254 cases of primary TNBC, HRD-positivity was associated with a better outcome after adjuvant chemotherapy for invasive disease-free survival, suggesting a more pronounced chemosensitivity in HRD-positive cancers in contrast to HRD-negative cases. In 67% of the HRD-positive cases, an identifiable molecular genetic alteration, presumably associated with the HRD-positive phenotype could be identified, e.g. BRCA1 or BRCA2 germline or somatic mutations, or BRCA1 promoter methylation (Johan Staaf, submitted). In addition, the occurrence of tumour infiltrating lymphocytes appears to be an important prognostic marker in ER-negative HER2-negative breast cancer (Mao, Qu et al 2016).

1.2 Background – chemotherapy in triple negative breast cancer (TNBC)

1.2.1 Standard adjuvant chemotherapy in triple negative breast cancer

Modern adjuvant chemotherapy in early HER2-negative breast cancer typically consists of a sequential anthracycline and taxane based therapy. Treatment is similar in ER-positive, and ER-negative disease. According to the Early Breast Cancer Trialists Collaborative Group (EBCTCG) metaanalysis the absolute added benefit of giving a taxane as part of the adjuvant treatment is in the order of 2% on survival at 8 years (EBCTCG 2012). In Europe many patients are treated with the regimen explored in the FNCLCC PACS01 trial that was evaluating the effect of the switch to three three-weekly cycles of docetaxel following three cycles of FEC100 (5-fluorouracil, epirubicine and cyclophosphamide), compared with a standard regimen consisting of six cycles of FEC100 (Roché, Fumoleau 2006). Different weekly, and three-weekly taxane regimens, following four cycles of adriamycin and cyclophosphamide (AC), have been compared in a large four-armed study indicating equivalent outcomes using three-weekly docetaxel and weekly paclitaxel, suggesting that either the three-weekly docetaxel regimen or the weekly paclitaxel regimen are the preferred options (Sparano, Wang 2008). The added benefit of the IV injection of 5-fluorouracil together with an anthracycline has been questioned, and is today omitted from standard treatment based on data from an Italian randomized trial (Del Mastro, De Placido 2015). A recently presented metaanalysis from the EBCTG strongly suggests that dose-dense chemotherapy given every two weeks decreases the disease recurrence rate, the 10-year breast cancer mortality and increases the overall survival compared with three-weekly treatment (EBCTCG 2017). In TNBC, a recent meta analysis including more than 2000 patients from nine individual phase 2 and 3 studies showed that the addition of platinum to preoperative treatment in early TNBC adds a benefit in terms of an increased pathological complete response (pCR) rate. The long term effect in terms of disease-free survival (DFS) and overall survival (OS) is still unclear, but based of the promising results the authors of the metaanalysis consider the addition of platinum an option in unselected TNBC (Poggio, Bruzzzone et al 2018).

In recent years, there has been an increasing interest to explore the advantage of studying treatment response in the preoperative (neoadjuvant) setting. Preoperative treatment of breast cancer generally leads to equal results in terms of DFS and OS compared with postoperative chemotherapy (Mieog, van der Hage et al. 2007; EBCTCG 2018). In a proportion of cases the preoperative chemotherapy treatment strategy leads to a clinically significant down-staging and a possibility of breast conservation that is not otherwise feasible. Equally importantly, the preoperative administration and response evaluation of preoperative medical treatment provides information regarding the prognosis of individual patients. In addition, it creates a unique opportunity to study the biological effects of treatment on site, and thus a possibility to assess the effect of chemotherapy in terms of response in individual patients and molecular genetic events in different and specific subgroups of breast cancer. Lastly, it is the fastest way of initiating medical treatment effective throughout the body, which is suggested of being of special importance in ER negative disease (Colleoni, Bonetti 2010; Gagliato de Melo, Gonzalez-Angulo 2014).

Although pCR has a strong prognostic value for the individual, the translation of pCR differences on a study level to differences in terms of event-free and OS, remains controversial (Cortazar P, Zhang L et al. 2014; Nekljudova V, Loibl S et al 2018). None the less, the prognostic value of a pCR differs between breast cancer subtypes, being greater among ER-negative breast cancers compared to ER-positive cases. In TNBC, the prognostic value of pCR/non-pCR is strong and has a potential of being a surrogate end-point for long-term outcome (Cortazar, Zhang et al. 2012; von Minckwitz, Untch et al. 2013). In addition to the prognostic value of the pCR/non-pCR itself, the properties of the residual cancer burden are associated with different outcomes, e.g. the rate of Ki67 positive cells in the residual tumour was prognostic for recurrence-free and overall survival in two studies (Symmans, Peintinger et al. 2007; Jones, Salter et al. 2009). Most importantly, preoperative trials gives us the potential to study response to treatment with the tumour in its original human environment and allows for the identification of biomarkers in the primary tumour predicting response or resistance to chemotherapy.

1.2.2 Carboplatin in early TNBC

The German GeparSixto trial investigated the effect of the addition of carboplatin in 595 HER2-positive and triple negative patients, randomized in two different strata. The primary endpoint was the difference in pCR rate with the addition to carboplatin to a polychemotherapy regimen consisting of paclitaxel and liposomal doxorubicin and bevacizumab for TNBC and trastuzumab + lapatinib for HER2-positive cases, and failed to show a statistically significant difference. However, a higher pCR rate was indeed observed in the triple negative subset associated with the addition of carboplatin, with rates of 38% without vs. 59% with the addition. The treatment regimen did show a significant toxicity only allowing 52% of the patients in the carboplatin-containing arm complete their preoperative chemotherapy (von Minckwitz, Schneeweiss et al. 2014).

The CALGB A 40603 included 443 TNBCs with clinical stage 2-3 (excluding inflammatory breast cancers) into a two-by-two randomized preoperative study adding carboplatin, bevacizumab, or both, to a chemo backbone consisting of paclitaxel q1w x 12 followed by AC q2w x 4. Carboplatin was given AUC6 x 4 together with paclitaxel. The addition of carboplatin was associated with a higher pCR rate in breast + axilla (41% vs. 54%) corresponding to an OR of 1,71; p=0.0029 (Sikov, Berry et al. 2015).

The I-SPY 2 randomized 72 + 62 HER2-negative breast cancer patients in a sophisticated adaptive design to standard treatment (paclitaxel q1w x 12 followed by AC q3w x 4) versus standard + carboplatin AUC6 q3w and the PARP-inhibitor veliparib 50 mg x 2 during the paclitaxel treatment. Among TNBCs the addition of carboplatin + veliparib resulted in a 26% to 51% increment in the pCR rate, indicating superiority with the addition of these drugs to standard treatment (Rugo, Olopade et al. 2016).

Recently, the BrighTNess trial was published. The study was designed following the I-SPY 2 results, and investigated the effect of the addition of carboplatin alone or in combination with the oral PARP-inhibitor veliparib to standard chemotherapy. In this important trial that included 952 patients randomized in a 1:1:2 manner to conventional taxane and anthracycline based treatment vs. treatment with the addition of carboplatin + placebo vs. carboplatin + veliparib. The primary end-point was pCR rate. pCR was significantly higher with the addition of carboplatin to standard chemotherapy, (58% vs. 31 %) but was not further increased with the addition of the combination of carboplatin and veliparib (53%) (Loibl S, O'Shaughnessy J, et al 2018).

The aforementioned metaanalysis by Poggio and coworkers included these four studies and an additional five trials investigating the effect on the pCR rate with the addition of carboplatin or cisplatin (one study only). The metaanalysis reported that the pCR rate was significantly higher with regimens including platinum (52.1%) compared with platinum free regimens (37.0%) (OR 1.96, 95% CI 1.46–2.62, P<0.001). Platinum based treatment increased the rate of grade 3-4 hematological toxicity, but the rate of neuropathy grade 3-4 was equal (Poggio, Bruzzone et al 2018).

1.2.3 Chemotherapy in BRCA-deficient and HRD-positive breast cancer

Attention has been drawn to the use of platinum salts in the treatment of BRCA deficient cancer as a way of exploiting the biological feature that is associated with homologous repair deficiency (Curigliano and

Goldhirsch 2011). In one preoperative study, the pCR rate after four cycles of cisplatin 75 mg/m² was 22% among 28 patients with stage 2-3 triple negative breast cancer. Signs of BRCA1-deficiency (low mRNA level, BRCA1 promoter methylation and BRCA1 germline mutation status) were predictive of a clinical response to treatment (Silver, Richardson et al. 2010). The pathological response rate in the breast and axilla was 62% in a set of 74 women treated after eight cycles of preoperative cisplatin, epirubicin, and paclitaxel (Frasci, D'Aiuto et al. 2010), and 61% among 104 patients with a deleterious BRCA1 germline mutation treated with four cycles of cisplatin 75 mg/m² (Byrski, Huzarsky et al 2014). In a Dutch study of high-dose chemotherapy, survival was significantly better after platinum- and thiotepa-based chemotherapy in cases with a comparative genomic hybridization classifier indicating BRCA-deficiency in the tumour (Vollebegg, Lips et al 2014). More recent studies have shown that not only BRCA germline mutations but also homologous repair deficiency in general is a predictor of response to conventional as well as platinum-based chemotherapy in triple negative and BRCA-associated breast cancer. In two small studies, Telli and co-workers have observed that HRD-positivity strongly predicts a pathologic complete response after conventional (Telli, Hellyer et al 2018) and platinum based chemotherapy (Telli, Timm et al. 2016). In our own research, we could observe that the effect of adjuvant chemotherapy was more pronounced in the HRD-positive subset compared with the HRD-negative breast cancers, giving support to the view that HRD is a marker of chemosensitivity in TNBC (Johan Staaf, submitted). Thus, it seems clear that markers of BRCA-deficiency and other indicators of homologous repair deficiency may be associated with an increased likelihood of response to platinum-based and conventional chemotherapy in subsets of TNBC.

1.2.4 Capecitabine in early breast cancer

Capecitabine is an oral prodrug to 5-fluorouracil, and an agent with a generally advantageous side-effect profile, and therefore frequently used in the palliative setting in breast cancer. Adding capecitabine to standard chemotherapy as an adjuvant treatment in early breast cancer has been exclusively evaluated in terms of long-term outcome in three large randomized trials of adjuvant treatment, and in one study of postoperative capecitabine after non-pCR to preoperative treatment. The FinXX study showed a non-significant but borderline benefit with adding capecitabine to docetaxel and EC at all cycles ($p = 0.087$, HR 0.79) (Joensuu, Kellokumpu-Lehtinen et al. 2009). In the TNBC sub-group consisting of 202 cases, however, the benefit appeared to be greater ($p = 0.018$, HR 0.48) (Lindman, Kellokumpu-Lehtinen et al. 2010). In the larger US Oncology study with similar design except less capecitabine was given (50% of cycles), the results were similar with a HR for DFS of 0.84 ($p=0.12$) and HR for OS of 0.68 ($p = 0.01$). Again, the results indicated a better relative effect in the TNBC sub-group (HR 0.81 for DFS and 0.62) and significant benefit in highly proliferative tumours (Ki67 > 10%, HR 0.70) (O'Shaughnessy, Paul et al. 2010; O'Shaughnessy, Koeppen et al. 2015). A study similar to the FinXX study is now being repeated in the TNBC sub-population in China (CBCSG10; ClinicalTrials.gov identifier NCT01642771). The study has included 585 cases of TNBC, treated adjuvantly with three cycles of docetaxel followed by three cycles of EC +/- capecitabine for the whole treatment course. At an interim analysis at 30 months of follow-up, recurrence free survival was numerically, however not significantly, advantageous in the capecitabine group corresponding to a HR of 0.57 (95% CI: 0.33-1.0).

In the Japanese-Korean CREATE-X/JBCRG04 trial, 910 HER2-negative breast cancer patients not achieving a pathological complete remission after conventional preoperative chemotherapy, were randomized to the addition of postoperative 6-8 cycles of capecitabine treatment versus no additional chemotherapy. A third of included patients were cases of TNBC. At the first planned interim analysis, DFS and OS was superior among patients receiving the additional capecitabine treatment (HR for OS = 0.59 95% CI 0.39-0.90; $p=0.01$). In the TNBC subset, the benefit in terms of OS of adding capecitabine corresponded to a HR of 0.52 95% CI 0.30-0.90 with an 8.5% absolute difference in survival at 5 years (Masuda, Lee et al 2017).

Finally, the Spanish CIOMBA/2004-01_GEICAM/2003-11 study investigated the effect of adding capecitabine to standard neo/adjuvant chemotherapy in TNBC. The total number of included patients in the study was 876 patients. DFS at five years of follow-up was 79,6 % with the addition of 8 cycles of capecitabine vs. 76,8 % without, corresponding to a HR of 0,82; $p=0,14$. However, in the subset of patients with a non basal-like phenotype, defined by the absence of cytokeratine 5/6 expression, there was a clinically relevant increased DFS and OS with the addition of capecitabine. The effect corresponded to a HR of 0,53 ($p=0,02$) with an absolute difference of 9,7 % (82,6 % vs. 72,9 %) between the study arms for DFS,

and a HR of 0,42 ($p=0,007$) with an absolute difference of 9,9 % (89,5 % vs. 79,6 %) for OS at five years of follow-up (Miguel, Barrios et al. 2018).

In brief, several pieces of evidence support a beneficial effect of capecitabine in the neo/adjuvant treatment setting in TNBC, but it is unclear whether this effect is attributable to unselected TNBC, or if it is restricted to a certain subset of cases. The Spanish CIOBMA-study suggests that the effect is preferential for the third of cases with a non-basal like TNBC phenotype. It should be noted that treatment schedules adding additional capecitabine after standard chemotherapy such as in the CREATE-X trial and the CIOBMA-study, leads to a protracted period of treatment for the patients, and that including it to the primary treatment would potentially be an advantage for the treated patients, provided equal effect, and that the subset with benefit could be identified.

1.3 Rationale for the Study and Purpose

The overall purpose of the NordicTrip study is to optimize chemotherapy in the curative treatment of early TNBC.

Standard adjuvant chemotherapy for unselected TNBC is currently based on the sequential use of anthracyclines with cyclophosphamide and taxanes. A recent metaanalysis has confirmed a beneficial effect in terms of pCR rate with the addition of carboplatin to conventional chemotherapy. A preliminary evaluation of the long-term outcome in the metaanalysis suggests a potential effect beyond response to platinum-based chemotherapy compared with conventional chemotherapy, with hazard ratios for DFS of 0.72 (95 % CI 0.49–1.06, $P=0.094$) and HR 0.86 for OS (95% CI 0.46–1.63, $P=0.651$). As a consequence, the authors state that platinum may be considered an option in unselected cases of TNBC (Poggio F., Bruzzone M, et al 2018). Based on these results we decided to include carboplatin in the baseline comparator arm in this trial, and to explore the potential additional effect of capecitabine in unselected and subsets of TNBC.

Based on subgroup analyses from three large trials, capecitabine appears to have an effect in unselected early TNBC. The CREATE-X, where capecitabine was given to patients following non-pCR to preoperative chemotherapy, has attracted the most attention. Here, RFS and OS was reported to be increased with the addition of capecitabine to standard chemotherapy with HRs of 0.58 and 0.52, respectively. The apparent benefit comes with the drawback of a protracted treatment period of almost a whole year on active chemotherapy, and indeed, the duration of capecitabine treatment in the study was reduced from 8 to 6 three-weekly cycles during the study. The effect of capecitabine treatment in the CREATE-X study is observed in a subset of patients with a relatively chemoresistant phenotype, i.e. cases that did not respond with a pCR on conventional chemotherapy (Masuda N., Lee SJ., et al. 2017). A fourth trial (the Spanish CIOBMA-study) suggests a benefit preferentially among non-basal-like TNBCs.

In TNBC, HRD has been observed to predict response to both conventional and platinum-based preoperative chemotherapy (Telli ML, Hellyer J., et al 2018; Telli ML, Timms KM, et al. 2016). However, the recent BrightNess trial failed to show an additional benefit by the addition of a PARP-inhibitor to platinum based preoperative chemotherapy (Loibl S, O'Shaughnessy J, et al 2018). We hypothesise that this might be an effect of a lack of further improved outcome in chemosensitive TNBC. Given the pronounced heterogeneity among TNBCs, it is reasonable to assume that the effect of capecitabine among TNBC may not be the same across different subsets of TNBC, and that it may primarily exert its effect in a relatively chemoresistant subset of TNBC.

The aim of the current study is to explore the effect of the addition of capecitabine in terms of pCR rate and long-term outcome to carboplatin-based preoperative chemotherapy in early TNBC, and to identify markers of response to treatment. We hypothesise that HRD, a marker of chemosensitivity in TNBC is a potential marker of response to the addition of capecitabine to platinum-based therapy in TNBC, with the most pronounced effect being seen in the HRD-negative subset. In addition, other pre-defined and exploratory biomarkers for response to treatment will be identified in the tumours and the blood and evaluated within the study.

In the study a marker of HRD-positivity (Oncoscan; Affymetrix) will be investigated as a potential predictor of response to preoperative treatment in the different treatment arms, and more specifically to the response of capecitabine.

2 STUDY OBJECTIVES AND RELATED ENDPOINTS

2.1 Primary and Secondary Endpoints

	Objectives	Endpoints
Primary	To compare the effect on pCR rate (ypTis, ypN0) of adding capecitabine to carboplatin-based preoperative chemotherapy in early ER-negative and HER2-negative breast cancer.	– pCR with the addition of capecitabine compared with carboplatin-based chemotherapy alone.
Primary translational	Evaluate HRD-status as a potential biomarker for response to the addition of capecitabine to carboplatin-based preoperative chemotherapy.	– Investigation of the potential interaction between the pCR-rate and treatment allocation based on stratification between HRD-positive and HRD-negative patients.
Secondary	To evaluate invasive disease free survival (IDFS), breast cancer specific survival (BCSS), distant recurrent free survival (DRFS) and overall survival (OS) with and without the addition of capecitabine. Toxicity and dose intensity in the treatment arms.	– IDFS, BCSS, DRFS and OS with the addition of capecitabine compared with carboplatin-based chemotherapy alone. – CTCAE (version 5.0) grade 3 to 5. – SAE. – Dose intensity for each drug.
Additional translational end-points	Comparing the pCR rate in the two treatment arms in different TNBC subtypes.	– pCR rate, IDFS, BCSS and OS in different subsets of molecularly characterized TNBC in the different treatment arms, this includes: – 1) Comparisons of patients with BRCA-deficiency vs. not – 2) Comparisons of patients with a germline vs. somatic vs. epigenetic loss of BRCA-function.

- 3) Comparisons of patients with different subtypes of TNBC based on gene expression profiling
- 4) Comparisons based on levels of markers of immune response (eg TILs and PDL1-expression).

3 OVERALL STUDY DESIGN

The study is a prospective, multicenter, open-label, phase III trial of preoperative chemotherapy with 1:1 randomization to one of two treatment arms.

Study Period Estimated date of first patient enrolled: 1 December 2019

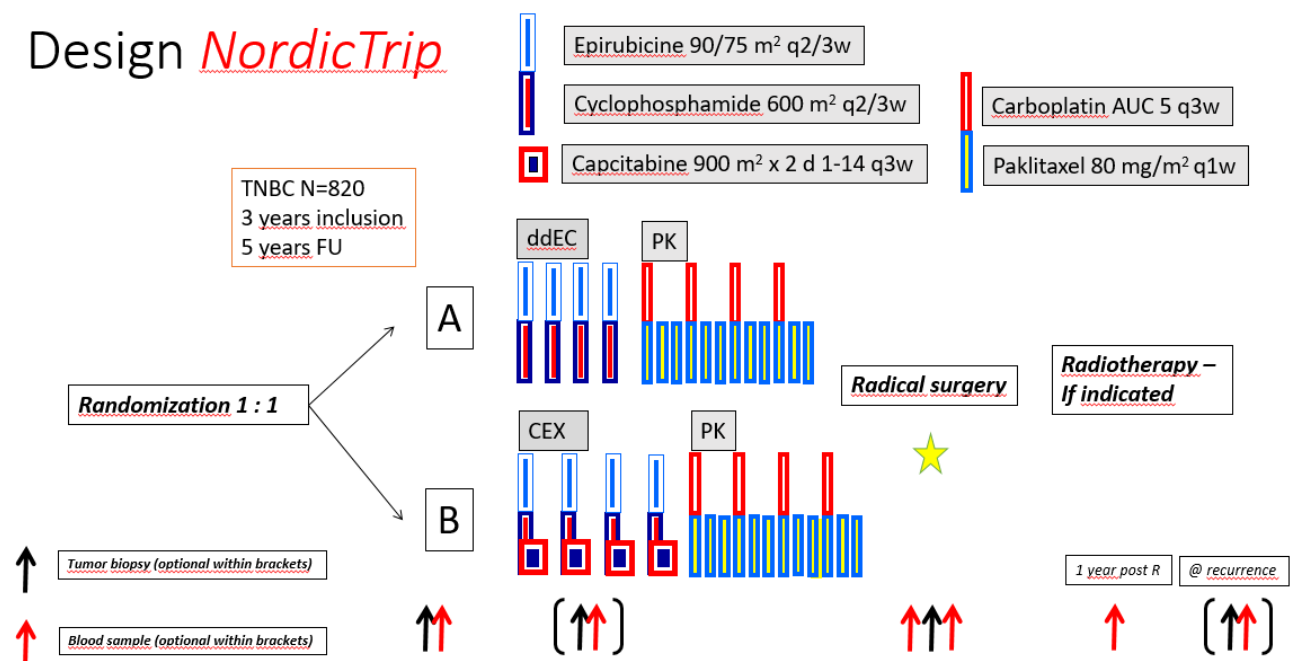
Anticipated recruitment period: 3 years

Estimated date of last patient completed treatment phase 30 June 2023

Treatment Duration: 20 - 23 weeks

Follow-up: 10 years (annual visits for 5 years and then yearly telephone contacts/visits up to 10 years)

Design *NordicTrip*



4 STUDY POPULATION

4.1 Selection of Study Population

Patients at participating hospitals that is planned for preoperative chemotherapy due to TNBC will be screened for inclusion in this study:

4.2 Number of Patients

820 patients will be included in this trial, see power calculation.

4.3 Inclusion Criteria

Patients must meet all of the following criteria to be eligible:

- 1) Signed written informed consent approved by the Ethical Review Board (IRB).
- 2) Age 18 to 75 years.
- 3) Histologically confirmed unilateral adenocarcinoma of the breast where neoadjuvant chemotherapy followed by definitive surgery is planned.
- 4) Node positive disease (N1-3) or if clinically N0 Tumor size ≥ 20 mm.
- 5) ER negative tumor defined by at least one the following:
 - a. ER < 1% cells positive by IHC
 - b. ER $\leq 10\%$ cells positive by IHC and PgR $\leq 10\%$ cells positive by IHC
- 6) HER2-normal tumor defined by at least one of the following:
 - a. HER2 0 or 1+ by immunohistochemistry
 - b. HER2 2+ by IHC and HER2 to centromere 17 ratio < 2.0 and/or less than 6 copies by in situ hybridization (ISH).
Comment: Patients with 3+ IHC for HER2 will not be included irrespective of the outcome of ISH-analysis.
- 7) Consent for germline mutation screening for BRCA1, BRCA2 and other inherited breast cancer associated genes.
- 8) WHO performance status 0 or 1.
- 9) Negative pregnancy test in women of childbearing potential (premenopausal or <12 months of amenorrhea post-menopause and who have not undergone surgical sterilization).
- 10) Willingness of female patients of childbearing potential, male patients and their sexual partners to use an effective means of contraception during the treatment period and at least 6 months thereafter.
- 11) Willingness by the patient to undergo treatment and study related procedures according to the protocol.

4.4 Exclusion Criteria

Patients meeting any of the following criteria are not eligible:

1. Clinical or radiological signs of metastatic disease.
2. Clinically inflammatory breast cancer
3. History of other malignancy within the last 5 years, except for carcinoma in situ of the cervix or non-melanoma skin cancer.
4. Previous chemotherapy for cancer or other malignant disease.

5. Charlson comorbidity index, excluding score for malignancy: (CCI) > 2, Comment: In patients 70-75 a CCI = 3 is allowed, see appendix B.
6. Inadequate organ function, suggested by the following laboratory results:
 - a Absolute neutrophil count < $1,5 \times 10^9/L$
 - b Platelet count < $100 \times 10^9/L$
 - c Haemoglobin < 90 g/L
 - d Total bilirubin greater than the upper limit of normal (ULN) unless the patient has documented Gilbert's syndrome
 - e ASAT (SGOT) and/or ALAT (SGPT) > $2,5 \times ULN$
 - f ASAT (SGOT) and/or ALAT (SGPT) > $1,5 \times ULN$ with concurrent serum alkaline phosphatase (ALP) > $2,5 \times ULN$
 - g Serum creatinine clearance < 50 ml/min
7. Concurrent peripheral neuropathy of grade 3 or greater (NCI-CTCAE, Version 5.0)
8. Patient who is actively breast feeding.
9. Assessed by the Investigator to be unable or unwilling to comply with the requirements of the protocol.

5 TREATMENT

Initial chemotherapy, see under 5.2, Dosage and drug administration.

After completion of chemotherapy, definitive surgery is performed followed by adjuvant radiotherapy according to local guidelines. Adjuvant postoperative bisphosphonates will be given according to local guidelines, but should not be initiated during the preoperative phase.

Patients that convert from a TNBC phenotype at baseline to a ER-positive or HER2-positive phenotype should receive additional adjuvant treatment as appropriate. Adjuvant capecitabine according to the CreateX-trial is not part of the study. In case additional adjuvant treatment is given, it should be thoroughly recorded in the CRF.

5.1 Drug Identity, Supply and Storage

All participating departments have experience of administration of epirubicin, cyclophosphamide, capecitabine, carboplatin and paclitaxel. Commercial formulations will be used and administered for all drugs. Known toxicities/side-effects are listed in Summary of Product Characteristics (SPC), see appendix C for the drugs used in the study. All investigational medicinal products (IMP) will be handled and administered according to clinical routines and local regulations.

5.2 Dosage and Drug Administration

Patients will be randomized between two treatment arms, A and B (see 10.2 Randomization).

A : **ddEC → PK**, ie: (epirubicin 90 mg/m^2 + cyclophosphamide 600 mg/m^2 q2w) x 4 followed by (paclitaxel 80 mg/m^2 day 1, 8, 15 + carboplatin AUC 5 day 1) x 4

B : **CEX → PK**, (epirubicin 75 mg/m^2 + cyclophosphamide 600 mg/m^2 + capecitabine 1800 mg/m^2 day 1-14) q3w x 4 followed by (paclitaxel 80 mg/m^2 day 1, 8, 15 + carboplatin AUC 5 day 1) x 4

Primary prophylaxis with G-CSF should be used in the ddEC regimen and might be used at the discretion of treating physician as part of the PK regimen.

5.2.1 Dosing of Arm A

Cycle 1-4: ddEC

- **ddEC** will be administered on day 1 of every 2-week cycle. Four cycles in total will be given. Regimen consists of epirubicin (E) 90 mg/m² as a 30 to 60-minute intravenous infusion followed by cyclophosphamide (C): 600 mg/m² as a 30-minute intravenous infusion. Epirubicin dose could be rounded to closest 5 mg and cyclophosphamide dose to closest 50 mg.
- **ddEC** doses should not be adjusted based on changes in body weight after first cycle given.
- **Hematopoietic growth factors (G-CSF)** G-CSF should be used daily day 4 to day 11 (treatment day is day 1) or, when using pegylated G-CSF, only given day 2. G-CSF is given SC and should be dosed based on patients weight according to local guidelines. G-CSF could be terminated before day 11 in case of neutrophils > 10.000 cells/mm³
- Patients aged 65-75 should start treatment on dose level -1, see table 1. For dose reduction recommendations see table 2 in section 5.4.

Cycle 5-8: PK

In arm A the first PK dosis will be administered three weeks after the last ddEC-dosis

- **Paclitaxel (P)** will be given at 80 mg/m² as a 60-minute intravenous infusion, on day 1, 8 and 15 at each 3 weekly cycle. Four cycles in total will be given. Doses could be rounded by 5 %. If treatment on day 8 or 15 is not given due to hematological toxicity, that treatment step is omitted. Paclitaxel doses should not be adjusted based on changes in body weight after cycle 1.
- **Carboplatin (K)** will be administered on day 1 of every 3-week cycle immediately after the infusion of paclitaxel. Regimen consists of carboplatin dosed based on Calvert formula at AUC (area under the curve) 5 as a 30-minute intravenous infusion. The carboplatin dose should not exceed 750 mg. Doses could be rounded to closest 10 mg. Calvert formula is based on patients GFR (glomerular filtration rate). GFR should be defined according to local guidelines. A total number of four doses should be given.
- Calvert Formula: Carboplatin dose in mg = Target AUC x (absolute GFR + 25)
- Patients aged 65-75 should start treatment on dose level -1, see table 1. For dose reduction recommendations see table 4 and 5 in section 5.4.

5.2.2 Dosing of Arm B

Cycle 1-4: CEX

- **EC** will be administered on day 1 of every 3-week cycle. Four cycles in total will be given. Regimen consists of epirubicin 75 mg/m² as a 30 to 60-minute intravenous infusion followed by cyclophosphamide: 600 mg/m² as a 30-minute intravenous infusion. Epirubicin dose could be rounded to closest 5 mg and cyclophosphamide dose to closest 50 mg.
- **Capecitabine (X)** will be taken as twice-daily 900 mg/m² oral doses (total daily dose of 1800 mg/m²) in the morning and evening in 3-week cycles consisting of 2 weeks of capecitabine treatment followed by 1 week without capecitabine treatment. Doses could be rounded to closest 150 mg. The first dose of each cycle will be administered as the evening dose on day 1 and the last dose of each cycle is scheduled the morning of day 15, followed by a 7-day rest period. This provides for a total of 28 single doses per cycle over 15 calendar days. The morning and evening dose should be given approximately 12 hours apart, and taken within 30 minutes after the ingestion of food with approximately 200 mL of water, ideally after the breakfast and evening meal.

- Doses should not be adjusted based on changes in body weight after the first CEX cycle.
- Patients aged 65-75 should start treatment on dose level -1, see table 1. For dose reduction recommendations section 5.4.2 and table 3.

Cycle 5-8: PK: see 5.2.1 above.

5.3 Premedication and Monitoring

5.3.1 Premedication ddEC Cycle 1-4 (Arm A)

An adequate antiemetic regimen should be given to prevent nausea and vomiting. G-CSF is mandatory on day 4-11, in case pegylated G-CSF is given it should be administered 24 h after chemotherapy.

5.3.2 Premedication CEX Cycle 1-4 (Arm B)

An adequate antiemetic regimen should be given to prevent nausea and vomiting. G-CSF should be avoided concomitantly with capecitabine treatment.

5.3.3 Paclitaxel premedication and precautions Cycle 5-8 (Arm A and B)

- Infusion bags, filters and sets should be PVC-free.
- **Cycle 5:** (Day 1, 8 and 15) paclitaxel is given as 60-minute infusion. The patient is monitored for hypersensitivity reactions with repeated blood pressure, pulse and general observation. In case of an infusion related reaction grade 1-2, stop infusion and treat according to local guidelines. Restart but de-escalate the infusion rate by 50 %. If no reaction occurs, the infusion rate may be increased again by 25 % after 15 minutes. If no reaction occurs during the first 30 minutes, the remaining amount of infusion is given at the regular rate.
- **Premedication** with corticosteroids, antihistamines and a H1 antagonist should be given according to local guidelines (e.g. oral doses of betamethasone 8 mg, clemastine 2 mg and ranitidine 150 mg 1-2 hours before paclitaxel).

Patients who experience a severe hypersensitivity reaction to paclitaxel should not be re-challenged with the drug, however nab-paclitaxel 100 mg/m² could replace paclitaxel at the discretion of the investigator.

5.3.4 Premedication Carboplatin Cycle 5-8 (Arm A and B)

An adequate antiemetic regimen should be given to prevent nausea and vomiting. Patients who experience a severe hypersensitivity reaction to carboplatin should not be re-challenged with the drug.

5.3.5 Monitoring of blood count

ANC, leucocytes and platelets should be measured ≤3 days before each treatment cycle.

5.4 Dose modifications

Patients will be monitored for toxicity during treatment. Dose adjustments will be performed according to dose levels in table 1 and the below guidelines for the drugs in the respective treatment arms. Patients aged 65-75 should start on dose level -1, see table 1.

Table 1 Dose reduction levels				
	Level 0 (start)	Level -1	Level -2	Level -3
Epirubicin A/B	90/75 mg/m ² *	75 /60 mg/m ²	60/50 mg/m ²	-
Cyclophosphamide	600 mg/m ²	500 mg/m ²	400 mg/m ²	-
Capecitabine (per dose)	900 mg/m ²	700 mg/m ²	500 mg/m ²	300 mg/m ²
Carboplatin	AUC 5	AUC 4	AUC 3	-
Paclitaxel	80 mg/m ²	65 mg/m ²	50 mg/m ²	-

5.4.1 Arm A and B, Cycle 1-4 (ddEC) dose reduction

Epirubicin and cyclophosphamide should be reduced together in case of severe hematological toxicity. In order to treat with ddEC on day 1, ANC should be at least $1.0 \times 10^9/L$ and platelets should be at least $100 \times 10^9/L$. If these values are not fulfilled, treatment is postponed until the values are achieved. Dose adjustments at the initiation of treatment are recommended as follows:

Dose levels are defined above in the table in chapter 5.4 and represent reductions of about 20%. Non-hematological grade 3-4 toxicity should result in dose reductions of 1-2 levels (20-40%) at the discretion of the treating physician.

Table 2 Dose reductions for dose dense epirubicin and cyclophosphamide (ddEC) based on hematological toxicity			
Neutrophils (x10⁹/L) at treatment day (22)	ddEC dose level	Platelets (x10⁹/L) at treatment day (22)	ddEC dose level
ANC ≥ 1.0	Stay on previous level	Platelets ≥ 100	Stay on previous level
ANC 0.5 – 0.9	Delay until ANC ≥ 1.0 and restart on previous level. On the second occasion reduce by one dose level	Platelets 50 – 99	Delay until Platelets ≥ 100 and restart on previous level. On second occasion: reduce by one dose level.
ANC < 0.5	Delay ddEC 1 week and reduce by one dose level. Restart when ANC ≥ 1.0 .	Platelets < 50	Delay until platelets ≥ 100 and reduce by one dose level.

Hematopoietic growth factors (G-CSF) G-CSF should be used day 4-11 (treatment day is day 1). In case pegylated G-CFS is given it should be administrated 24 h after chemotherapy. G-CSF is given SC and should be dosed based on patients weight according to local guidelines.

5.4.2 Arm B Cycle 1-4 (CEX) dose reduction

General remarks regarding capecitabine treatment:

Capecitabine therapy might result in severe hand-foot syndrome or diarrhea in about 20-25 % of treated patients and caution is important. Other non-hematological toxicity could occur as well. Reduced renal function and old age are risk factors. The nurse or treating physician should check patients for adverse events at a weekly basis in cycle 1-2, and thereafter, if toxicity is manageable only in day 8 and before next cycle.

- If grade 2-4 toxicity occurs during the 14 days of ongoing capecitabine treatment, interrupt capecitabine treatment until next cycle.
- If greater than grade 1 non-hematological toxicity persists at the time for next cycle (day 22), treatment should be delayed until the toxicity has resolved to grade ≤ 1 before starting next cycle and reduce by one dose-level according to table 1.

Epirubicin and cyclophosphamide should be reduced together in case of severe hematological toxicity. Capecitabine should be reduced alone based on capecitabine related toxicities (diarrhea, hand-foot). Dose levels are defined at table 1 and represent reductions of about 20%. Non-hematological grade 3-4 toxicity should result in dose reductions of 1-2 levels (20-40%) at the discretion of the treating physician.

Table 3 Dose reductions for cyclophosphamide, epirubicin and capecitabine (CEX) based on hematological toxicity. In case of hematological toxicity only, the capecitabine dosis should not be altered.			
Neutrophils (x10⁹/L) at treatment day	CEX dose level	Platelets (x10⁹/L) at treatment day	CEX dose level
ANC ≥ 1.0	Stay on previous level	Platelets ≥ 100	Stay on previous level
ANC 0.5-0.9	Delay until ANC ≥ 1.0 and restart on previous level. On the second occasion: reduce by one dose level.	Platelets 50 – 99	Delay until Platelets ≥ 100 and restart on previous level. On second occasion: reduce by one dose level.
ANC < 0.5	Delay until ANC ≥ 1.0 and reduce by one dose level.	Platelets < 50	Delay until platelets ≥ 100 and reduce by one dose level.

5.4.3 Arm A and B, Cycle 5-8 dose reduction

Based on partly different toxicity profiles, delivered doses of carboplatin and paclitaxel may be adjusted individually. Moderate to severe thrombocytopenia should lead to a dose reduction of carboplatin while neurotoxicity should lead to a dose reduction of paclitaxel according to Arm A table 5. In case of neutropenia the addition of G-CSF is recommended. If toxicity cannot be attributed to one of the drugs, both drugs could be reduced one or two steps according to treating physician's discretion. If carboplatin is discontinued before cycle 4, the last cycle/-es should be given with only paclitaxel. If paclitaxel is discontinued before cycle 4 the last cycle/-es should be given with only carboplatin, in these cases re-escalation of carboplatin should be considered up to a maximal dosis corresponding to AUC = 5. However, if both drugs are discontinued, omitted cycles could be replaced by EC (i.e. if cycle 4 is omitted due to toxicity, one additional course of EC could be given to a total of 5 cycles of EC).

In the case of severe hypersensitivity reactions to paclitaxel or carboplatin, the drug should be stopped and discontinued for remaining cycles. Paclitaxel could be replaced by nab-paclitaxel 100 mg/m² if hypersensitivity is related to paclitaxel. Long lasting bone-marrow depression is the dominating toxicity of carboplatin and the table below is to be used at each treatment course for dose reductions, delays or the addition of G-CSF.

Carboplatin hematological toxicity: In order to treat with carboplatin on day 1 ANC should be at least 1.0 x 10⁹/L and platelets should be at least 100 x 10⁹/L. If these values are not fulfilled, treatment is postponed until the values are achieved. Dose adjustments at the initiation of treatment are recommended as follows:

Table 4 Arm A and B: Dose reductions of Carboplatin day 1 based on hematological toxicity			
Neutrophils (x10⁹/L) at treatment day	Carboplatin dose	Platelets (x10⁹/L) at treatment day	Carboplatin dose
ANC ≥ 1.0	Stay on the same dose level	Platelets ≥ 100	Stay on the same dose level
ANC < 1.0	Postpone treatment until ANC ≥ 1.0 and stay on the same dose level, add G-CSF	Platelets <100	Postpone treatment until platelets ≥ 100 and reduce by one dose level

Hematopoietic growth factors (G-CSF) may be used to reduce the length and severity of neutropenia and thereby reducing the risk for febrile neutropenia and treatment delay. G-CSF is to be used daily on day 3-5, 10-12 and 17-19 (treatment day is day 1) and should be dosed based on patients weight according to local guidelines.

Table 5 Arm A and B: Dose reductions for Paclitaxel based on neurotoxicity (CTCAE grades)	
Cycle 1 Paclitaxel	Cycle 2-4 Paclitaxel, day 1
If no sign of neuropathy give: 80 mg/m²	If no neurotoxicity give 80 mg/m²
If prevalent grade 1 neurotoxicity give: 65 mg/m²	If grade 1 neurotoxicity give 65 mg/m²
If prevalent grade 2 neurotoxicity give: 50 mg/m²	If grade 2 neurotoxicity give 50 mg/m²
If prevalent grade 3-4 neurotoxicity: patient cannot be included in the study	If grade 3-4 neurotoxicity discontinue paclitaxel

Paclitaxel hematological toxicity: In order to treat with paclitaxel on day 8 and 15 ANC should be at least $1.0 \times 10^9/L$ Platelets should be at least $75 \times 10^9/L$. If these values are not fulfilled, that weekly paclitaxel treatment is omitted. No other action is taken. At the second occasion, either the addition of G-CSF or a dose reduction by one step is considered at the next treatment day.

5.5 Concomitant Medication

At study initiation, patients should continue with their concomitant medications, as directed by their physician. Any cancer therapeutic or surgical procedure performed during the study period should be recorded in the CRF including the date, indication, description of the procedure(s) and any clinical findings.

5.5.1 Supportive Care

Supportive care for disease-related symptoms will be offered as needed to all patients in this study.

5.5.2 Hematopoietic Growth Factors

Hematopoietic growth factors (G-CSF) may be used to treat symptomatic neutropenia as well as prophylactically according this protocol chapter 5.4 or based on local guidelines or responsible physicians' choice. The use of G-CSF should be registered in the eCRF as well as in the patient's medical records. The use of hematopoietic growth factors should be avoided concomitantly with capecitabine treatment.

5.5.3 Other Supportive Measures

Patients should not routinely receive prophylactic antibiotics during the study.

5.5.4 Oral Coumarin-Derived Anticoagulants

Patients receiving concomitant capecitabine or oral coumarin-derived anticoagulants should have their anticoagulant response (INR or prothrombin time) monitored frequently in order to adjust the anticoagulant dose accordingly. Altered coagulation parameters and/or bleeding, including death, have been reported in patients taking capecitabine concomitantly with coumarin-derived anticoagulants such as warfarin and phenprocoumon. A PK interaction has been observed. The use of low molecular weight heparin instead of coumarin during chemotherapy is at the discretion of the investigator but is strongly recommended.

5.5.5 Phenytoin

Increased phenytoin plasma concentrations have been reported during concomitant use of capecitabine and phenytoin. Formal drug-drug interaction studies with phenytoin have not been conducted. Patients

taking phenytoin concomitantly with capecitabine should be regularly monitored for increased phenytoin plasma concentrations and associated clinical symptoms.

5.5.6 Allopurinol

Interactions with allopurinol have been observed for fluorouracil; with possible decreased efficacy of fluorouracil. Concomitant use of allopurinol with capecitabine should be avoided.

5.5.7 Antivirals and Antiprotozoals

Capecitabine should not be administered together with the antiviral drug sorivudine or its chemically related analogues, such as brivudine. A clinically significant drug-drug interaction between sorivudine and fluorouracil, resulting from the inhibition of DPD by sorivudine, has been described in the literature. This interaction, which leads to increased fluoropyrimidine toxicity, is potentially fatal.

Metronidazole increased the toxicity of fluorouracil in patients with colorectal cancer, apparently by reducing the clearance of the antineoplastic. As it has been described in the literature, caution should be exercised.

5.5.8 Gastrointestinal Drugs

Pretreatment with cimetidine for 4 weeks led to increased plasma concentrations of fluorouracil following intravenous and oral administration in six patients. The effect was probably due to a combination of hepatic enzyme inhibition and reduced hepatic blood flow. No such effect was seen following single doses of cimetidine in five patients or pretreatment for just one week in six. Care is required in patients taking both drugs simultaneously.

5.5.9 Other Anticancer Therapies

The use of other cytotoxic agents, investigational drugs, breast cancer or immunotherapy (other than that specified by the protocol), are not allowed during the study treatment period.

The use of GnRH-analogues for the purpose of preservation of ovarian function in premenopausal patients is allowed during the treatment phase of the study.

Patients requiring radiotherapy during study to other body sites than the breast and the regional lymphatic nodes will be considered to have had relapsed and should discontinue study before the start of radiotherapy. Such patients will be evaluated based of intention to treat and will be considered to not having fulfilled criteria for pCR.

Patients converting from a triple negative phenotype in the pretherapeutic biopsy to an ER-positive or HER2-positive phenotype at radical surgery should receive endocrine and/or HER2-directed treatment as appropriate. If such treatment is considered it should be thoroughly reported in the CRF.

5.6 Subject Compliance

The timing and doses of IMPs used in this trial are to be recorded in the CRFs. For Capecitabine, the dosis and treatment interval should be recorded. Relative given dose will be calculated for each IMP.

6 STUDY PROCEDURES

6.1 Subject Numbering

Each subject is identified in the study by a unique subject number that is assigned when the subject is randomized in the study. Once assigned the subject number cannot be reused for any other subject.

6.2 Study Phases

6.2.1 Screening phase

The Screening Phase is the time between the date a patient provides written informed consent, and the date of randomization. Data collection and procedures during this time period include patient demographics, eligibility requirements, concomitant medications, medical history, physical examination/vital signs including assessment of WHO performance status, adverse events, serious adverse events, laboratory measurements, pregnancy testing (if applicable), biospecimens (blood and tissue samples), evaluations of tumor burden, report from sentinel node biopsy procedure. CT of thorax and abdomen is mandatory in high risk patients with clinical lymph node positive disease or T3-4 disease. All procedures should be performed within 28 days prior to randomization See table 6. Pregnancy test (when applicable) must be performed within 14 days prior to randomization.

6.2.2 Treatment phase

Study treatment should begin within 14 days after randomization. During the treatment phase, data collection and procedures include the following: assessment of physical examination/vital signs, WHO performance status, laboratory measurements including mutation analyses for hereditary breast cancer genes, adverse events, serious adverse events, treatment adherence (interruptions, dose modifications and termination), any postoperative chemotherapy, and determination of recurrence status.

Collection of biomaterial from the primary tumor site before and after study treatment, and blood samples for study specific analyses in the baseline, as well as pre- and postoperatively is mandatory. Further material collection of biomaterial (see table 6), blood samples and tissue from recurrence (loco-regional, distant metastasis or a contralateral breast cancer) is optional.

6.2.3 Follow-up phase

During Follow-Up, patient visits should be performed yearly until 5 years after randomization. Visit dates are calculated from the date of randomization. During year 6-10, yearly telephone contacts should be performed and documented by the study nurse or investigator.

6.3 TABLE 6 SCHEDULE OF ASSESSMENTS AND PROCEDURES:

Assessment procedures:	Screening phase		Treatment phase			Before surgery	At surgery	End-of-treatment visit	Post-Therapy Follow-up	Post-Therapy Follow-up
	Completed no more than (time) prior to randomization		Cycle 1 day 1	Before treatment cycle: 2, 4, 5, 6, 7 & 8	Before treatment cycle: 3			1-4 weeks after surgery	1-5 years after randomization (Annual +/- 4 weeks)	6-10 years after randomization (Annual +/- 4 weeks)
Informed consent (a)	X	28 days								
Inclusion/exclusion criteria	X	14 days								
Demographics, medical history (b)	X	14 days								
Cancer/treatment history (c)	X	28 days								
Physical examination incl vital signs (d)	X	14 days			X	X		X	X	
Clinical assessment of locoregional tumor extent	X	14 days		If clinically indicated before c5 (o)	X	X				
Blood sample routine (7 ml) Hematology; Biochemistry (e)	X	14 days		X	X					
Pregnancy test (f)	X	14 days		X	X			X		
Serum estradiol and FSH	X	14 days						X	X (@ year 1 and year 2)	
Determination of renal clearance (g)				Before cycle 5						
Blood sample 7 ml for mutation screening of germline alterations associated with an increased breast cancer risk or homologous repair deficiency, Clinical analysis (h)			X							
Blood sample (2 x 7 ml) for research purpose (i)			X		X					

Assessment procedures:	Screening phase		Treatment phase			Before surgery	At surgery	End-of-treatment visit	Post-Therapy Follow-up	Post-Therapy Follow-up
	Completed no more than (time) prior to randomization		Cycle 1 day 1	Before treatment cycle: 2, 4, 5, 6, 7 & 8	Before treatment cycle: 3			1-4 weeks after surgery	1-5 years after randomization (Annual +/- 4 weeks)	6-10 years after randomization (Annual +/- 4 weeks)
Blood sample (10 ml) for the study of circulating tumour markers (j)			X		X	X		X	X (year 1 only)	
ECG	X	28 days								
CT thorax/abdomen (k)	X	28 days								
Collection of FFPE tissue from primary tumour (l)	X	28 days			X (optional)		X			
Collection of tumour tissue in RNA later (optional) (m)	X	28 days			X		X		In case of recurrence (optional)	In case of recurrence (optional)
Breast imaging Mammography/US, or MRI (MRI optional) (n)	X	28 days		If clinically indicated before c5	X	X				
Axillary staging, ie Fine needle aspiration (cN+) and/or SN biopsy in cN0 disease (o)	X	28 days					X			
Study drug compliance				X	X	X				
Registration of AE (study nurse or treatment nurse) (p)			X	X	X	X				
Survival follow-up / additional cancer therapies (q)									X	X
Screening mammography (according to local policies)									X	X
Quality of life assessment (r) (Selected patients only)	X	14 days						X	X (@ year 1 and year 2)	
Telephone contact										X

- a. Written informed consent must be obtained before any study-specific screening procedures are performed
- b. Includes date of birth and menopausal status

- c. Includes data on current breast cancer (i.e. date of diagnosis, histological type, histological grade, ER/PgR-receptor status, HER-2 status, Ki-67).
- d. Includes a clinical examination, WHO performance status (see Appendix E), height (baseline only), weight, pulse rate and blood pressure and clinical TNM-status. Preoperative visit may be performed at the department of surgery.
- e. Hematology (haemoglobin, white blood cell (WBC) count, absolute neutrophils and platelet count) and serum chemistry (ASAT/ALAT, alkaline phosphatase, sodium, potassium, total calcium, total bilirubin, serum creatinine and albumin.) will be sampled within 14 days prior to randomization and within 3 days prior to the start of subsequent chemotherapy cycles.
- f. Pregnancy test (P-hCG or U-hCG) is required within 14 days prior to randomization for all women of childbearing potential (menstruating women or women with menstruation within 12 months of study entry), during the entire treatment phase, and 1-4 weeks after radical surgery. In the case surgery is not performed, a pregnancy test should be performed at the corresponding time point about 5-6 weeks after the last chemotherapy cycle.
- g. GFR should be defined according to local guidelines.
- h. Whole blood (7 ml) for screening for germline alterations associated with an increased breast cancer risk is sent to the BRCA-lab Avd för onkologi, Lunds Universitet, SE-223 81 Lund, Sweden
- i. Blood sample (2 x 7 ml) used for the analysis of circulating immune markers, and as a reference to differentiate between germline and somatic genetic alterations in tumor tissue.
- j. Blood samples (10 ml at each occasion) for the assessment of circulating tumour DNA and other circulating tumor and immune markers will be collected from all patients at Cycle 1 Day 1, Cycle 3 Day 1, at the end of Treatment Phase (before and after surgery), at 1 year after randomization and at the time of recurrence.
- k. Screening for metastatic disease is mandatory in clinical N+ disease or T3 disease. In the case of N+ disease at pretherapeutic sentinel node biopsy only, such screening may be omitted. All staging studies including physical exam and CT must show no evidence of metastatic disease, including suspicious lymphadenopathy or skin nodules on physical exam. The first hand choice is a CT scan with iv and oral contrast. Other staging studies are at the treating physician's discretion or in a situation where a CT is contraindicated. In these cases eg MRI of the thorax, a ultrasound examination of the liver and a bone scan may replace a CT scan. Any other staging test being performed as clinically indicated (e.g. CT scans, MRI, ultrasound of abdomen, PET scans) must be negative for metastatic disease.
- l. At least four ultrasound guided core needle biopsies collected in formalin are sent to the tissue coordinating centre. Two cores is the absolute minimum, four cores are preferred and strongly encouraged. Two cores each will be used for pathologic evaluation and biomolecule extraction respectively. At surgery: In case of pCR a sample from the tumor bed is taken. In case of non-pCR a sample of the residual tumor.
- m. Optional core needle biopsies (N=1-2) collected in RNA later collected at base line and after 2 treatment cycles.
- n. All patients should undergo breast imaging after two treatment cycles using the same methods as during screening. In the case of a poor or questionable response after two treatment cycles, extra imaging and clinical evaluation could be done after four treatment cycles, ie before the change of treatment phases.
- o. The assessment of N-stage is performed according to local routines, but fine needle aspiration of clinically suspicious axillary lymph nodes (cN1) is mandatory. If cytologically negative cN1, or in cN0, a sentinel node biopsy before start of treatment or at radical surgery is recommended.
- p. Adverse events are collected during study treatment by the responsible physician, treatment nurse or study nurse before each treatment and reported in the CRF.
- q. New cancers or additional cancer therapies documented for all patients until recurrence or for a minimum of 5 years following randomization. All patients should at least have annual visits until 5 years post-randomization and phone contact for additional 5 years. If consented by the patient, tumor tissue is collected from metastatic tissue at time of recurrence, as well as blood (10 ml) for the analysis of circulating tumor DNA and other circulating tumor and immune markers.
- r. In a subset of patients, a quality of life assessment will be performed using validated instruments during screening, after surgery and at the visits one and two years post randomization.

6.4 Procedures and Investigations

For specific timepoints, see section 6.3, Table 6. Study Flow Chart.

Informed consent

Informed consent must have been given voluntarily by each subject before any study specific procedures are initiated.

Clinical status and vital signs

At screening, a physical examination including WHO performance status, a palpation of breast/chest wall, axillae, supra- and infraclavicular region, cardio-pulmonary examination, blood pressure, pulse rate, height and weight is required. Symptom-directed physical examinations, cardio-pulmonary examination, blood pressure, weight and pulse rate will be performed at subsequent visits. Data on the medical history including disease history and corresponding treatment details is registered.

Concomitant medication

During follow-up, information on all breast cancer related therapies must be recorded.

Tumour evaluation

- CT or MRI scan of thorax/abdomen as screening for metastatic disease in cases with clinically node positive disease (N+ by sentinel node biopsy only excluded) or clinically T3-4 tumors.
- Breast imaging (mammography and ultrasound and/or MRI) before and after two treatment cycles and according to local routines during follow-up. In case of uncertain response, a second on treatment evaluation can be added after four treatment cycles.
- Clinical assessment of the locoregional tumor extent will be recorded during screening, after two treatment cycles and preoperatively after the end of preoperative chemotherapy. For details see 7.1.
- Patological assessment of the breast (and axilla) at time of surgery. For details see 7.1.

Laboratory analysis

- It is recommended that the following assessments be at the same laboratory for each patient. The frequency of assessments is provided in the schedule of assessments (Table 6). All initial laboratory assessments during the screening period must be performed within 14 days prior to randomization.
- Hematology: haemoglobin, white blood cell (WBC) count, absolute neutrophils and platelet count.
- Blood chemistry with liver function tests: ASAT/ALAT, alkaline phosphatase, sodium, potassium, total calcium, total bilirubin, serum creatinine and albumin.
- Plasma/Serum/urine pregnancy test for women of child-bearing potential (within 14 days before randomization).
- Serum Estradiol and FSH.
- Additional hematology/chemistry panels will be performed as clinically indicated.
- Determination of glomerular filtration rate (GFR) must be done prior to start of carboplatin treatment (see section 5.2.1)

Study specific parameters – biospecimens, for details see protocol for translational research.

- Collection of blood (10 ml) for the analysis of inherited breast cancer genes, including BRCA1 and BRCA2, this analysis will be performed as a clinical analysis where all clinically relevant result will be reported back to the study site for documentation in the CRF and will be shared with the patient. The reported gene panel will be defined based on what is considered clinical practice at the participating study sites, and will generally follow national guidelines.

- Collection of blood (10 ml) for study specific analysis of germline alterations in order to be able to differentiate somatic from germline alterations when analyzing the tumor tissue, these analyzes will not be reported back to the patient.
- Collection of blood samples for ctDNA and other circulating tumor markers at c1d1 (10 ml), after two treatment cycles, after completed chemotherapy, after radical surgery and one year post randomization. In addition, an optional blood sample may be collected and at tumor recurrence.
- Collection of FFPE tissue from primary tumour at baseline including two to four ultra-sound guided core needle biopsies of the primary tumor for study specific biomarker analysis. Four cores are preferred and at least two are needed. Preferably two (but at least one) cores will be used for pathologic analyses, and preferably two (but at least one) for biomolecule extraction.
- Collection of FFPE tissue after two treatment cycles preferably two cores, but a minimum of one core (optional).
- Collection of FFPE tissue at surgery, either from residual disease or from the tumor bed in the case of a complete remission (required).
- Collection of tumour tissue in RNA later at baseline, after two treatment cycles and at surgery (optional, one core/tissue specimen at each occasion)

6.4.1 By visit

Screening phase

- Collection of Informed Consent.
- Inclusion/exclusion criteria assessment.
- Demographics, Medical history including data on current breast cancer, and assessment of prevalent symptoms using NCI common Terminology Criteria for Adverse Events, CTCAE Version 5.0.
- Physical examination.
- Baseline mammography and ultrasound (and/or MRI) of the affected breast.
- Fine needle aspiration of clinically suspicious axillary lymph nodes.
- Sentinel node biopsy in case of cN0 disease, including cytologically negative enlarged nodes will be performed according to local praxis.
- Marking of the tumor area eg using charcoal or a coil will be performed according to local routines, and is strongly encouraged.
- CT or other metastatic screening for patients with cN+ or cT3-4.
- Baseline assessment of routine blood analyses.
- Pregnancy test for women of childbearing potential.
- Serum Estradiol and FSH.
- ECG
- Collection of FFPE tissue from primary tumor, and collection of tumor tissue in RNA-later (optional).
- In a subset of patients, quality of life assessment.

Treatment phase

- C1d1: Clinical screening for germline mutations in inherited breast cancer genes including BRCA1 and BRCA2 and other germline alterations that are in clinical use.
- At d1 of each cycle: Toxicity assessment using NCI common Terminology Criteria for Adverse Events, CTCAE Version 5.0 (may be registered by the study nurse).
- Physical examination, including locoregional status of the breast and lymphnodes before c3 and after the completion of chemotherapy. In the case of a poor response to treatment after two cycles, if clinically indicated an extra tumor evaluation should be initiated after the completion of the first treatment phase (four treatment cycles).
- Research blood samples at c1d1 and before c3.
- Routine blood analyses, including pregnancy test for women of childbearing potential before each treatment cycle (7 ml).
- Determination of renal clearance before the carboplatin treatment phase.

- Breast imaging (mammography and ultrasound and/or MRI) before c3 and before surgery. In the case of a poor response to treatment after two cycles, if clinically indicated an extra tumor evaluation should be initiated after the completion of the first treatment phase.
- Ultrasound guided core needle biopsies after two treatment cycles will optionally be collected according to the same protocol as during the screening phase.
- Before c3 (optional) collection of FFPE tumor tissue and tumor tissue in RNA-later.
- Study drug compliance.

Before surgery

To be performed after completion of chemotherapy (could be done at the department of surgery).

- Physical examination, and overall assessment of clinical treatment response: Clinical response (ClinR), Stable disease (SD), and Progressive disease (PD) (This assessment is meant to correspond to RECIST 1.1 criteria although RECIST is not defined primarily for mammography and ultrasound of breast tumors. ClinR will be used for a situation corresponding to complete response and partial response according to RECIST 1.1, see appendix F.).
- Breast imaging.
- Research blood sample before surgery (circulating tumor markers).
- Study drug compliance.
- Toxicity assessment using NCI common Terminology Criteria for Adverse Events, CTCAE Version 5.0.

At surgery

- Tissue collection for the study of residual disease.
- Axillary staging according to local guidelines.

End of Treatment visit

- Collection of a research blood sample.
- Physical examination.
- Pregnancy test.
- Serum Estradiol and FSH.
- Blood sample for the study of circulating tumor markers.
- In a subset of patients, quality of life assessment.

Post therapy follow-up

- Annual follow-up visits for 5 years followed by annual phone contacts for an additional 5 years, the telephone contacts could be replaced by physical visits.
- Physical exam year 1-5 after randomization.
- Screening mammography during 10 years according to local guidelines.
- Serum Estradiol and FSH, at the visits one and two years after randomization.
- Blood sample for the study of circulating tumor markers, year 1 after randomization.
- Other investigations as appropriate if the patient presents with symptoms suggesting a recurrence. In a subset of patients, quality of life assessment, at the visits one and two years after randomization.

6.5 Procedures for Discontinuation

6.5.1 Patient Discontinuation

Patients who withdraw or are withdrawn from the study, will stop further treatment but will be followed up regarding potential outcome of radical breast surgery, recurrence, and death according to what the patient consents to.

If possible, a final assessment will be made (end of study visit). The reason for discontinuation will be recorded. If the patient makes a partial withdrawal of consent this should be thoroughly recorded in the medical record by the treating physician. The investigator is obliged to follow up any significant adverse events until the outcome either is recovered or resolved, recovering/resolving, not recovered/not resolved, recovered/resolved with sequelae, fatal or unknown.

Patient's who withdraw or are withdrawn from the study before start of treatment, will be replaced.

6.5.2 Criteria for Patient Discontinuation

Patients may be discontinued from study treatment and assessments at any time. Specific reasons for discontinuing a patient for this study are:

- Voluntary discontinuation by the patient who is at any time free to discontinue his/her participation in the study, without prejudice to further treatment.
- Safety reason as judged by the Principal Investigator
- Major protocol deviation
- Incorrect enrollment ie, the patient does not meet the required inclusion/exclusion criteria for the study
- Patient lost to follow-up
- Pregnancy (only applicable during the treatment phase of the study)
- Disease progression
- Patient's serious non-compliance to study treatment and/or procedures

Patients that discontinue study participation will be evaluated according to intention-to- treat (patients not undergoing definitive surgery will be evaluated as having a non-pCR for the primary endpoint).

6.5.3 Trial Discontinuation

The whole trial may be discontinued at the discretion of the sponsor in the event of any of the following:

- Occurrence of AEs unknown to date in respect of their nature, severity and duration
- Medical or ethical reasons affecting the continued performance of the trial
- Difficulties in the recruitment of patients

The sponsor will inform all investigators, the relevant Competent Authorities and if applicable Ethics Committees of the termination of the trial along with the reasons for such action. If the study is terminated early on grounds of safety, the Competent Authorities and Ethics Committees will be informed within 15 days.

7 ASSESSMENTS

7.1 Assessment of Efficacy

The following parameters of response will be recorded:

- Clinical evaluation and imaging of the locoregional tumor extent during screening, before c3 and before radical surgery. In case of an uncertain response before c3, additional clinical and imaging-based evaluation may be performed as clinically indicated before c5 and should in those cases be recorded in the CRF. This evaluation is primarily for security purpose in order to exclude the

possibility of progressive disease. The result before three, (five) and before surgery will be evaluated in three categories based on imaging and clinical examination: Clinical response (ClinR), Stable disease (SD), and Progressive disease (PD) (This assessment is meant to correspond to RECIST 1.1 criteria (Eisenhauer et al.) although RECIST is not defined primarily for mammography and ultrasound of breast tumors. ClinR will be used for a situation corresponding to complete response and partial response according to RECIST 1.1), see appendix F.

- Pathological evaluation at time of radical surgery. Here a pathological complete response is defined as the complete disappearance of signs of invasive breast cancer cells in the breast and locoregional lymph nodes. Residual insitu deposits are allowed. It should be noted that a patient that has undergone a negative sentinel node biopsy at baseline, or a sentinel node biopsy with only signs of microscopic disease (Nmic) does not have to undergo further axillary surgery provided there is a clinical response in the breast. In the case such a patient does not show signs of invasive disease in the breast at surgery, he or she is considered to have a pathological complete response.
- Optionally response will also be reported according to Miller-Payne or Residual Cancer burden criteria.
- Assessment of tumour recurrence when clinically indicated: CT/MRI scans of thorax and abdomen and bone scintigrams. In addition, CT/MRI scans must be obtained for anatomic regions not covered by the thorax and abdomen scans, in subjects where there is clinical suspicion for deep soft tissue metastases. If possible use the same methods as at baseline in patients subjected to distant metastases screening. Histologic verification of metastatic disease during follow-up is strongly encouraged.

7.2 Safety and Tolerability Assessments

Safety will be monitored by the assessments described below as well as the collection of AEs at every treatment visit. For details on AE collection and reporting, refer to Section 8.

For the assessment schedule refer to Flow chart, see table 6.

Clinical status and vital signs

Physical examination including WHO performance status, a palpation of breast/chest wall, axillae, supra- and infraclavicular region, cardio-pulmonary examination, blood pressure, pulse rate, height (screening only) and weight is required.

Laboratory evaluation

Local laboratory will be used for the analysis of hematology and biochemistry:

- Hematology: haemoglobin, white blood cell (WBC) count, absolute neutrophils and platelet count.
- Blood chemistry with liver function tests: ASAT/ALAT, alkaline phosphatase, sodium, potassium, total calcium, total bilirubin, serum creatinine and albumin.
- Serum Estradiol and FSH
- Additional hematology/chemistry panels will be performed as clinically indicated.

7.3 Quality of life (QoL)

A subset of included patients (N=250) will be evaluated for QoL during screening, at the end of chemotherapy, at the visit one year after randomization and two years after randomization.

8 SAFETY MONITORING AND REPORTING

The investigator is responsible for the detection and documentation of events meeting the criteria and definition of an adverse event (AE) or serious adverse event (SAE).

The methods for collection of safety data are described below.

After the inclusion of 60 patients of the study, a thorough data analysis will be performed in order to evaluate the safety, tolerability and feasibility of the study regimens. If necessary an amendment to the protocol will be submitted, including adjustments in the treatment schedules and dosing recommendations.

8.1 Definitions

8.1.1 Adverse Event (AE)

An AE is any untoward medical occurrence in a patient administered a pharmaceutical product and which does not necessarily have a causal relationship with this treatment.

An adverse event (AE) can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of a medicinal (investigational) product, whether or not related to the medicinal (investigational) product.

The term AE is used to include both serious and non-serious AEs.

For this particular study, only AEs grade 3-5 (CTCAE, version 5.0) will be registered and reported from the signing of the informed consent and until 30 days after the last dose of study drug. However, for sensory and motor neuropathy toxicity grade 1-5 should be reported in order to adjust paclitaxel treatment doses.

Disease progression, breast cancer recurrences and/or death due to breast cancer are excluded from reporting AEs/SAEs. If an abnormal laboratory value/vital sign are associated with clinical signs and symptoms, the sign/symptom should be reported as an AE and the associated laboratory result/vital sign should be considered additional information that must be collected on the relevant CRF.

8.1.2 Serious Adverse Event (SAE)

Any untoward medical occurrence that:

- Results in death
- Is immediately life-threatening
- Requires in-patient hospitalization or prolongation of existing hospitalization
- Results in persistent or significant disability or incapacity
- Is a congenital abnormality or birth defect
- Is an important medical event that may jeopardize the subject or may require medical intervention to prevent one of the outcomes listed above.

Medical and scientific judgment is to be exercised in deciding on the seriousness of a case. Important medical events may not be immediately life-threatening or result in death or hospitalization, but may jeopardize the subject or may require intervention to prevent one of the listed outcomes in the definitions above. In such situations, or in doubtful cases, the case should be considered as serious. Hospitalization for administrative reason (for observation or social reasons) is allowed at the investigator's discretion and will not qualify as serious unless there is an associated adverse event warranting hospitalization.

For this particular study, planned hospitalization in association with breast cancer surgery, treatment and other study procedures, including sentinel node biopsy will not be reported as SAEs.

8.1.3 Suspected Unexpected Serious Adverse Reactions (SUSAR)

SUSAR is the term used to refer to an adverse event that occurs in a clinical trial subject, which is assessed by the sponsor and/or study investigator as being unexpected, serious and as having a reasonable possibility of a causal relationship with the study drug. To meet the criterium unexpected the AE nature or severity could not be listed in the applicable product information.

8.2 Disease Progression/Recurrence

Events which are definitely due to disease progression will not be reported as an AE/SAE. However, if the investigator considers that there was a causal relationship between treatment with IP or protocol design/procedures and the disease progression/recurrence, then this must be reported as an SAE.

Death due to progressive disease is to be recorded on a specific form in the CRF but not as an SAE.

Any new primary cancer (non-related to the cancer under study) should be considered for reporting as an SAE.

8.3 Time Period for Reporting AE and SAE

For each patient the standard time period for collecting and recording AE and SAEs will begin at the start of the screening phase (signature of informed consent), through the treatment phase and will continue for at least 30 days following the last dose of study treatment for each patient.

During the course of the study all AEs and SAEs will be proactively followed up for each patient; events should be followed up to resolution, unless the event is considered by the investigator to be unlikely to resolve due to the underlying disease. Every effort should be made to obtain a resolution for all events, even if the events continue after discontinuation/study completion. Note that during the screening phase, only SAEs due to procedures stated by the protocol and not performed routinely will be reported, Eg an allergic reaction to intravenous contrast infusion associated with a CAT scan that would have been performed as part of clinical routine.

8.4 Recording of Adverse Events

If the patient has experienced adverse event(s), the following information will be recorded in the CRF:

- The nature of the event(s) will be described in precise standard medical terminology (i.e. not necessarily the exact words used by the patient).
- The duration of the event will be described in terms of event onset date and end date.
- The intensity of the adverse event: Grade according to Common Terminology Criteria for Adverse Events version 5.0 (CTCAE).
- The causal relationship of the event to the study medication will be assessed as one of the following:

Not Related:

There is not a temporal relationship to investigational product administration (too early, or late, or investigational product not taken), or there is a reasonable causal relationship between non-investigational product, concurrent disease, or circumstance and the AE.

Possibly Related:

There is reasonable causal relationship between the investigational product and the AE. Dechallenge information is lacking or unclear.

Related:

There is a definite causal relationship between the investigational product and the AE. The event responds to dechallenge.

- Action taken
- The outcome of the adverse event – whether the event is resolved or still ongoing.

8.5 Reporting Procedure

8.5.1 AEs and SAEs

All adverse events and serious adverse events that should be reported as defined in section 7.2 will be recorded in the patient's CRF. SAEs must be reported by the investigator to the sponsor/coordinating centre, Clinical Research Unit, KFE Dept. of Hematology, Oncology and Radiation Physics, Skåne University Hospital, SE-205 02 Malmö, Sweden, email: SAE.onkologi@skane.se, within 24 hours after the site has gained knowledge of the SAE. In case of an SAE the Serious Adverse Event Report Form must be completed, signed and sent by e-mail to the coordinating centre. The initial report shall promptly be followed by detailed, written reports if necessary. The initial and follow-up reports shall identify the trial subjects by unique code numbers assigned to the latter.

The sponsor keeps detailed records of all SAEs reported by the investigators and performs an evaluation with respect to seriousness, causality and expectedness.

8.5.2 SUSARs

SUSARs will be reported to the Competent Authority and Ethics Committee according to national regulation. The following timelines should be followed:

The sponsor will ensure that all relevant information about suspected serious unexpected adverse reactions that are fatal are reported to the Competent Authority within seven days, for other SUSARs the corresponding time limit is 15 days after knowledge by the sponsor of such a case, and that relevant follow-up information is subsequently communicated within an additional eight days.

All other suspected serious unexpected adverse reactions will be reported to the Competent Authority concerned and to the Ethics Committee concerned as soon as possible but within a maximum of fifteen (15) days of first knowledge by the sponsor.

SUSARs will be reported to the sponsor using the CIOMS-form by the site and to the respective competent authority in the country of the site, according to applicable regulations in each country.

8.5.3 Annual Safety Report

Once a year throughout the clinical trial, the sponsor will provide the Competent Authority with an annual safety report. The format will comply with national requirements.

8.6 Procedures in Case of Emergency

The investigator is responsible for assuring that there are procedures and expertise available to cope with emergencies during the study.

8.7 Data Monitoring Committee (DMC)

The conduct of the study and will be supervised by an independent data monitoring committee that will ensure patient safety and that the study is carried out according to ICH-GCP and the Declaration of Helsinki (October 2013), in the best interest of the patients. The committee will perform a toxicity analysis after the inclusion and treatment of the first 60 patients.

8.8 Contraconception and pregnancy testing

Participating patients with childbearing potential, and male patients should actively avoid pregnancy using a highly effective means of birth control during the treatment period and at least 30 days thereafter. Study treatment could only be initiated after a negative pregnancy test no more than 14 days prior to randomization. Pregnancy tests will be performed before each treatment cycle and 1-4 weeks after radical surgery. Highly effective means of contraception include the use of an intrauterine device, surgical sterilization (bilateral tubal ligation in females or bilateral vasectomy in males), or heterosexual abstinence. In theory, also combined hormonal and progestogen-only contraception associated with inhibition of ovulation are classified as a highly effective means of birth control, but these are not recommended in association with breast cancer.

The use of progestogen-only oral hormonal contraception where inhibition of ovulation is not the primary mode of action, male or female condom with or without spermicide and cap, diaphragm or sponge with spermicide are birth control measures that are *not considered as highly effective*. This includes the combined use of male condom with either cap, diaphragm or sponge with spermicide (double barrier methods)

9 DATA MANAGEMENT AND MONITORING

9.1 Case Report Form (CRF)

The designated investigator staff will enter the data required by the protocol into the electronic case report forms (eCRF). The Principal Investigator is responsible for assuring that data entered into the eCRF is complete, accurate, and that entry is performed in a timely manner. The signature of the investigator will attest the accuracy of the data in each eCRF. If any assessments are omitted, the reason for such omissions will be noted on the eCRFs. Corrections, with the reason for the corrections will also be recorded.

9.2 Source Data

The medical records for each patient should contain information, which is important for the patient's safety and continued care, and to fulfill the requirement that critical study data should be verifiable.

To achieve this, the medical records of each patient should clearly describe at least:

- That the patient is participating in the study, e.g. by including the enrollment number and the study code or other study identification;
- Date when Informed Consent was obtained from the patient and statement that patient received a copy of the signed and dated Informed Consent and version.
- Results of all assessments confirming a patient's eligibility for the study;
- Diseases (past and current; both the disease studied and others, as relevant);
- Surgical history, as relevant;
- Results of assessments performed during the study;
- WHO performance status assessments conducted as part of the study;

- Treatments given, changes in treatments during the study and the time points for the changes;
- Visits to the clinic / telephone contacts during the study, including those for study purposes only;
- Serious Adverse Events (if any) including causality assessments;
- Adverse Events will be reported in a work sheet (source data) at each treatment cycle if needed, and relevant parts will be described in the medical record;
- Date of, and reason for, discontinuation from study treatment;
- Date of, and reason for, withdrawal from study;
- Date of death and cause of death, if available;
- Additional information according to local regulations and practice.

9.3 Study Monitoring

The investigator will be visited by the Clinical Study Monitor, who will check the following: Informed consent process, reporting of adverse events and all other safety data, adherence to protocol, maintenance of required regulatory documents and source data verification (SDV).

The Data Manager will regularly review the relevant eCRFs for completeness and a risk based monitoring plan will be performed meaning that centers with better completeness and accuracy will have fewer visits by the study monitor.

The study monitor looks for accuracy and will ask the site staff to adjust any discrepancies as required.

When the data manager has checked and verified the eCRFs, the database will be locked for further handling and statistical evaluation.

Sponsor's representatives (e.g. monitors, auditors) and/or competent authorities will be allowed access to source data for source data verification in which case a review of those parts of the hospital records relevant to the study may be required.

9.4 Confidentiality

The investigator shall arrange for the secure retention of the patient identification and the code list. Patient files shall be kept for the maximum period of time permitted by each hospital. The study documentation (CRFs, Site File etc) shall be retained and stored during the study and for 15 years after study closure. All information concerning the study will be stored in a safe place inaccessible to unauthorized personnel.

9.5 Database management

The data reported in this trial will be collected in the data base held by the Danish Breast Cancer Group. The eCRF will provide range check for relevant data and the data base will regularly be reviewed by the Data Manager. Queries will be sent to centers in case of unclear or missing data. Statistical calculations as part of the risk based monitoring plan will be performed to optimize accuracy. At time for the feasibility analysis, interim analyses and completion of the study statistical analyses will be performed after cleaning and locking of the data.

10 STATISTICAL METHODS AND DATA ANALYSIS

10.1 Determination of Sample Size, and randomization.

For statistical analysis plan see Appendix G.

Primary aim: Pathological complete response rate after preoperative chemotherapy is the primary end-point of the study, which will be evaluated by comparing the effects of neoadjuvant administration of a carboplatin-based treatment (A) and treatment adding capecitabine (B) on pCR. By assuming a pCR rate of 52% (Poggio F., Bruzzone M. et al. 2018) in arm A, a sample size of 808 patients, 404 in each arm, is required to have 80% power to detect a difference of 10 percentage units, i.e. a pCR rate of 62% in arm B, using a two-sided Chi-squared test with the significance level set at 5%. To account for an expected low fraction of non-evaluated patients, the sample size is set at 820. Patients that receive study treatment, but are not operated will be evaluated as having a non-pCR.

Primary translational aim: To investigate if the effects of the treatments depend on HRD-status. More specifically, the aim is to test for differential effect of the two treatments on pCR for HRD-negative (HRD low and intermediate by oncoscan) and HRD-positive (HRD high by oncoscan) patients.

Assuming that 59% of the patients are HRD-positive (implying that 41% are HRD-negative) and that the overall pCR rate is 52% in treatment arm A, the pCR rate in HRD- is given by $(0.52 - 0.59 \cdot p)/0.41$, where p is the pCR rate in HRD+ for treatment A. Hence, this pCR rate can be calculated for a series of reasonable values of p : for example a pCR rate in HRD+ of 63% implying a pCR rate of 36% in HRD- corresponding to an OR for HRD+ vs HRD- in treatment A of 3.0. or an even larger effect, e.g. 67% vs 30% implying an OR of 4.6. If we, furthermore, assume the same effect of the treatments A and B in HRD+ (OR 1.0) and that the overall pCR rate is 62% in treatment arm B, the pCR rate in HRD- in arm A is given by $(0.62 - 0.59 \cdot p)/0.41$. Hence, the OR for treatment B vs A can be calculated. To sum up, the set of assumptions:

- 59/41 ratio HRD+/HRD-
- 52% pCR in arm A
- 62% pCR in arm B
- OR = 1.0 for pCR when comparing B vs A in HRD+

Imply that the OR for pCR when comparing B vs A in HRD- can be calculated for each possible value of the pCR rate in e.g. HRD+ for treatment A.

Holding the sample size fixed at approximately 800 patients in total for the two groups, the power to detect differential effect of B vs A on pCR in HRD+ and HRD- using a two-sided test of interaction at the 5% significance level can be calculated. Below follows two examples:

- 63% pCR in HRD+ (for both arms) implies 36.2% pCR in HRD- for arm A and 60.6% pCR in HRD- for arm B corresponding to an OR of 2.71 for B vs A in HRD-, an effect which can be detected with 92% power with the chosen sample size
- 67% pCR in HRD+ (for both arms) implies 30.4% pCR in HRD- for arm A and 54.8% pCR in HRD- for arm B corresponding to an OR of 2.77 for B vs A in HRD-, an effect which also can be detected with 92% power with the chosen sample size

10.2 Randomization

Randomization will be performed 1 : 1 between the two treatment arms A and B, with stratification for the following factors:

- 1) Clinical T stage. Comment: If several imaging modalities are available, these should be considered in the following order: MRI followed by ultrasound and mammography. T4 inflammatory disease is evaluated based on clinical appearance.
- 2) Clinical N stage, Comment: the cN0 stratum includes patients without imaging-based signs of nodal involvement that undergo a sentinel node biopsy irrespective of the result of the pathological assessment of the sentinel node. Comment: This is in order to allow for randomization also in the event that the pathological evaluation of the SN is not yet available at time of randomization.
- 3) Country

11 STUDY MANAGEMENT

11.1 Investigator Delegation Procedure

The principal investigator is responsible for making and updating a “delegation of tasks” listing all the involved co-workers and their role in the project. Performing tasks according to clinical praxis does not define that particular individual as a “co-worker” in the project, eg chemotherapy administration. The PI will ensure that appropriate training relevant to the study is given to all co-workers in the study, and that any new information of relevance to the performance of this study is forwarded to the staff involved.

11.2 Protocol Adherence

Investigators ascertain they will apply due diligence to avoid protocol deviations.
All significant protocol deviations will be recorded and reported in the Clinical Study Report (CSR).

11.3 Study Amendments

If it is necessary for the study protocol to be amended, the amendment and/or a new version of the study protocol (Amended Protocol) must be notified to and approved by the Competent Authority and the Ethics Committee according to EU and national regulations.

11.4 Audit and Inspections

Authorized representatives of a Competent Authority may visit the centre to perform inspections, including source data verification. Likewise the representatives from sponsor may visit the center to perform an audit. The purpose of an audit or inspection is to systematically and independently examine all study-related activities and documents to determine whether these activities were conducted, and data were recorded, analyzed, and accurately reported according to the protocol, Good Clinical Practice (ICH GCP), and any applicable regulatory requirements. The principal investigator will ensure that the inspectors and auditors will be provided with access to source data/documents.

12 ETHICAL AND REGULATORY REQUIREMENTS

The study will be conducted according to the Declaration of Helsinki and ICH-GCP and all applicable regulatory requirements. Registration of patient data will be carried out in accordance with national personal data laws.

12.1 Ethics Committee Approval

The study protocol, including the patient information and informed consent form to be used, will be approved by the ethics committee in each participating country before enrolment of any patients into the study.

The coordinating investigator in each country is responsible for informing the ethics committee of any serious and unexpected adverse events and/or major amendments to the protocol.

12.2 Other Regulatory Approvals

The protocol will be submitted and approved by the Medical Product Agency in each country before commencement of the study.

The protocol will also be registered in www.clinicaltrials.gov before inclusion of the first patient.

12.3 Informed Consent Procedure

The investigator is responsible for giving the patients verbal and written information about the nature, purpose, possible risk and benefit of the study. They will be informed as to the strict confidentiality of their patient data, but that their medical records may be reviewed for trial purposes by authorized individuals other than their treating physician.

It will be emphasized that the participation is voluntary and that the patient is allowed to refuse further participation in the protocol whenever she/he wants. This will not prejudice the patient's subsequent care. Documented informed consent must be obtained for all patients included in the study before they are registered in the study. This will be done in accordance with the national and local regulatory requirements. The investigator is responsible for obtaining signed informed consent.

A copy of the patient information and consent will be given to the patients. The signed and dated patient consent forms will be filed in the Investigator Site File binder.

12.4 Subject Identification

The investigator is responsible for keeping a list of all patients (who have received study treatment or undergone any study specific procedure) including patient's date of birth and personal identification number and full names.

The patients will be identified in the eCRFs by patient number, initials and date of birth.

12.5 Risk-benefit analysis

Treatment arm A includes dose-dense EC, a treatment that has been extensively studied in clinical trials, and that is proven more effective in a recent meta-analysis compared with conventionally scheduled regimens. In treatment arm B, the oral agent capecitabine is added to the EC treatment. Due to the anticipated toxicity profile, capecitabine and EC will be given with an interval of three weeks, keeping in mind that the oral capecitabine per se is a dose dense treatment given twice daily for two out of three weeks. After four treatment cycles, both study arms contain a combined carboplatin-taxane part, consisting of iv carboplatin AUC 5 three-weekly in parallel with weekly paclitaxel x 12. The decision to include carboplatin in the comparator arm was made since a recent metaanalysis comparing preoperative treatment in approximately 2000 patients with TNBC with and without the addition of carboplatin has shown a significant and relevant improvement in the rate of patients that experience an optimal response to preoperative chemotherapy with the addition of carboplatin. There is a substantial experience in the clinic from this type of treatment that is expected to result in an increased hematological toxicity during the treatment period, in comparison with a non-carboplatin containing arm. The carboplatin containing treatment arm, has been and is being investigated in several other studies, including treatment studies in metastatic breast cancer. Also the CEX treatment has been extensively evaluated in previous studies.

Two measures are taken in order to minimize the risk of inadequate treatment toxicities in any of the study arms. 1) a rigorous dose adjustment scheme is included in the protocol, allowing for dose adjustments based on experienced individual toxicity in the different study arms. 2) An interim analysis focusing on treatment toxicity after 60 treated patients, whereafter potential adjustment in the study regimens will be made with the aim of reasonable and equivalent toxicity between the study arms.

Two general ethical issues may be seen in a study of this design; undertreatment and overtreatment. Both problems are relevant, but largely unavoidable in curative medical treatment in breast cancer, where

a fraction of patients will recur in spite of intense treatment efforts, and others would not have had a recurrence even if they would not have had any medical treatment at all. This is applicable even in the adverse subset of TNBCs. Undertreatment might primarily be an issue in the comparator arm. We have dealt with this using an optimal and modern treatment regimen including a dose dense treatment step, and with the addition of carboplatin, which we consider to be a tolerable regimen given the seriousness of the disease. It is our judgement that the carboplatinbased treatment arm meets current standards of a modern adjuvant treatment regimen in early TNBC. Overtreatment, ie excessively toxic treatment, may be an issue in both treatment arms, just as it is in any adjuvant medical treatment in breast cancer where a fraction of patients would in fact be cured by surgery alone. However, the problem of over treatment is not associated to the study itself, but is equal in the population of patients treated outside of this trial. All patients would receive treatment even outside of the study. Still, if the study is negative, the added toxicity of capecitabine, would occur without benefit to the individual patients. Given the fact that this added toxicity is generally reversible, we consider this risk ethically acceptable.

13 END-OF-STUDY

Recruitment will stop after the enrollment of 820 patients. The study will be formally closed when the last patient has finished the last follow-up, 10 years after start of treatment according to table 6. The actual time-point depends on when the last patient will actually be included in the study. The study may be prematurely discontinued at the discretion of the sponsor as described in section 6.5.3. A Declaration of End-of-Trial Notification will be sent to the competent authorities within 90 days from the End-of-Study. Safety and efficacy data will be reported to the EudraCT database within 12 months from the End-of-Study date.

14 TRIAL SPONSORSHIP

Department of Hematology, Oncology and Radiation Physics, Skåne University Hospital is the sponsor of the trial.

15 TRIAL INSURANCE

Swedish patients treated within this trial are covered by patientskadeförsäkringen.

Danish patients treated within this trial are covered by patienterstatningen (www.pebl.dk).

Finnish patients treated within the trial are covered by the national patient insurance (Potilasvahinkovakuutus).

16 PUBLICATION POLICY

Upon study completion and finalization of the study report the results of this study will either be submitted for publication and/or posted in a publicly assessable database of clinical study results. Results from the feasibility analysis, the neoadjuvant part and interim analysis will be reported on international meetings and could be submitted for publication.

The results of this study will also be submitted to the Competent Authority and the Ethics Committee according to EU and national regulations.

All personnel who have contributed significantly with the planning and performance of the study (Vancouver convention 1988) may be included in the list of authors.

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18 LIST OF APPENDICES

A List of participating centres

B. Charlson comorbidity index (ref Charlson et al 1987)

C. Summary of Product Characteristics (SmPC):

D: **Common Terminology Criteria for Adverse Events (CTCAE) Version 5-0**, Published: November 27 2017, US department of Health and Human Services, National Institute of Health, National Cancer Institute, https://ctep.cancer.gov/protocoldevelopment/electronic_applications/docs/ctcae_v5_quick_reference_8.5x11.pdf

E: WHO performance status

F: Recist criteria

G: Statistical Analysis Plan

Appendix A – List of participating centers

15 sites in Sweden:

Region Skåne SUS (Lund/Malmö and Helsingborg)

Region Halland (Halmstad)

Region Kronoberg (Växjö)

Region Väst (Gothenburg, Borås)

Region Stockholm (Södersjukhuset and Capio St Görans sjukhus)

Region Uppsala-Örebro (Uppsala, Örebro, Gävle, Västerås, Karlstad)

Region Sydost (Jönköping)

Region Norr (Umeå, Sundsvall)

8 sites in Denmark

3 sites in Finland

1 site in Iceland

APPENDIX B – CHARLSON COMORBIDITY INDEX

(REF CHARLSON ET AL 1987)

Age	<50 years	0
	50-59 years	+1
	60-69 years	+2
	70-79 years	+3
	≥80 years	+4
Diabetes Mellitus	None	0
	Uncomplicated	+1
	End-organ damage	+2
Liver disease	None	0
	Mild	+1
	Moderate-severe	+3
Malignancy (not applicable in this study)	None	0
	Any leukemia, lymphoma or localized solid tumor	+2
	Metastatic solid tumor	+6
AIDS	No	0
	Yes	+6
Moderate to severe Chronic Kidney Disease	No	0
	Yes	+2
Congestive Heart Failure	No	0
	Yes	+1
Myocardial Infarction	No	0
	Yes	+1
Chronic Obstructive Pulmonary Disease	No	0
	Yes	+1
Peripheral Vascular Disease	No	0
	Yes	+1
Cardiovascular Accident or TIA	No	0
	Yes	+1
Dementia	No	0
	Yes	+1
Hemiplegia	No	0
	Yes	+2
Connective Tissue Disease	No	0
	Yes	+1
Peptic Ulcer Disease	No	0
	Yes	+1

APPENDIX C – Summary of Product Characteristics (SPC)

Capecitabine: https://www.ema.europa.eu/documents/product-information/capecitabine-accord-epar-product-information_en.pdf

Carboplatin:

https://docetp.mpa.se/LMF/Carboplatin%20Accord%20concentrate%20for%20solution%20for%20infusion%20SmPC_09001be68022e1b3.pdf

Cyclophosphamid:

https://docetp.mpa.se/LMF/Sendoxan%20powder%20for%20solution%20for%20injection%20SmPC_09001be680015ef6.pdf

Epirubicin:

https://docetp.mpa.se/LMF/Epirubicin%20Teva%20solution%20for%20injection%20or%20infusion%20SmPC_09001be68022da56.pdf

Paclitaxel:

https://docetp.mpa.se/LMF/Pacligen%20concentrate%20for%20solution%20for%20infusion%20SmPC_09001be68028cde7.pdf

Appendix D – Common Terminology Criteria for Adverse Events (CTCAE) Version 5-0:

Published: November 27 2017, US department of Health and Human Services, National Institute of Health, National Cancer Institute,
https://ctep.cancer.gov/protocoldevelopment/electronic_applications/docs/ctcae_v5_quick_reference_8.5x11.pdf

Appendix E – WHO performance status:

Grade	Explanation of activity
0	Fully active, able to carry on all pre-disease performance without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work
2	Ambulatory and capable of all selfcare but unable to carry out any work activities. Up and about more than 50% of waking hours
3	Capable of only limited selfcare, confined to bed or chair more than 50% of waking hours
4	Completely disabled. Cannot carry on any selfcare. Totally confined to bed or chair
5	Dead

Appendix F – Resist criteria and adoption for clinical response evaluation in NordicTrip

<https://project.eortc.org/recist/wp-content/uploads/sites/4/2015/03/RECISTGuidelines.pdf>

Clinical response will be based on breast imaging and will be classified and recorded according to the following principles: Clinical response (ClinR), Stable disease (SD), and Progressive disease (PD). This assessment is meant to correspond to RECIST 1.1 criteria (Eisenhauer et al.), although RECIST is not defined primarily for mammography and ultrasound of breast tumors. ClinR will be used for a situation corresponding to complete response and partial response according to RECIST 1.1).

Appendix G – Statistical analysis plan

STATISTICAL ANALYSIS PLAN

The primary objective of the study is to compare pathologic complete response (pCR) rates in the two treatment groups. The intention-to-treat (ITT) principle will be followed and a one-degree-of-freedom chi-square test with significance threshold α set to 0.05 will be used to determine significance. The pCR rates will be reported as well as the absolute difference between these rates with 95% confidence interval (CI). Furthermore, the relative risk of non-pCR, with 95% CI will be estimated using a univariable generalized linear model (glm) with log-link.

The primary translational objective, which is to determine if the relative treatment effect varies with Homologous Repair Deficiency (HRD) status (positive vs negative/intermediate), will also be evaluated in a glm with log-link. This model will include main effects for the dichotomous variables randomization group and HRD status and the interaction between these two variables. Statistically significant differential treatment effect by HRD status will be claimed if the p-value for the interaction term is below 0.05.

pCR rates will also be compared in subgroups of patients, for example by HRD-status, PDL1-status and in subgroups defined by BRCA-status: germline BRCA-mutation, somatic BRCA-mutation, promoter methylated BRCA and no BRCA-aberration.

The above mentioned analyses will be carried out as soon as possible after trial has been closed for inclusion and the data on pathological outcome at the time of radical surgery is available, ie at the earliest time about 6 months after the inclusion of the last patient.

The secondary objectives related to long-term survival (invasive disease free, breast cancer specific, distant recurrence free and overall survival) will be evaluated in relation to randomization group, again adhering to the ITT principle. Standard methods for survival analysis will be used, i.e. Kaplan-Meier estimates, log-rank tests and Cox-regression analysis. Focus will be on estimated relative effects with 95% CI from Cox-regression analyses and not on p-values. We foresee a problem with interpretation of the analyses of treatment effects on long term survival due to an expected clinical use of postoperative adjuvant therapy after evaluation of pCR among patients not achieving a pathological complete remission. We have reason to believe that future clinical guidelines will suggest postoperative capecitabine treatment as an option in triple negative breast cancer patients not having achieved a pathologic complete response at radical surgery, following preoperative chemotherapy. We cannot estimate the extent of such additional treatment in the study cohort, but it will most probably be restricted to patients not having a pathological complete response. Sensitivity analyses will be carried to investigate the magnitude of this problem, e.g. by ignoring postoperative adjuvant chemotherapy (ITT) or by censoring the follow-up at the date of starting postoperative chemotherapy.

The timing of the survival analyses depends on the event rates. Simulation show that, assuming uniform inclusion of 2x410 patients over three years, exponentially distributed survival times in both randomization groups and a 5-year survival of 70% for the treatment with the poorest survival, 300 events is necessary to have 80% power to detect a hazard ratio of 0.70 for the other group relative to the chosen reference group with a log-rank test with significance level set to 0.05. But, again, focus will not be on p-values in these analyses, but on relative effects with 95% CI:s. Over the 1000 simulations, under these assumptions mentioned above, the timing of event number 300 varied from 3 to 6 years after inclusion of the last patient.

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